

机器人机构学 (Robotic Mechanisms)

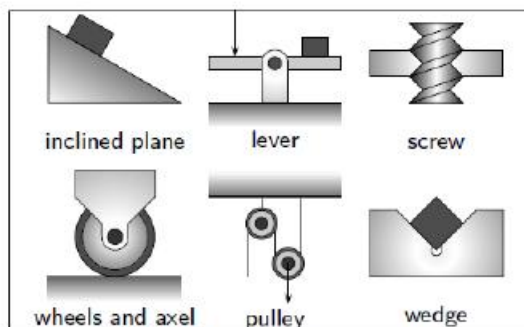
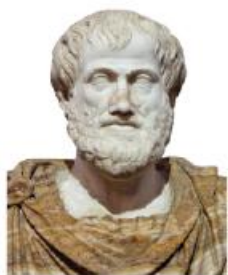
Chapter 1. 机器人与机构学基础

1.1. 机器人与机构学的发展历史与研究内容

1. 机器人与机构学基础

1.1. 机器人与机构学的发展历史与研究内容

- Ancient History (Before 1450 A.D.)

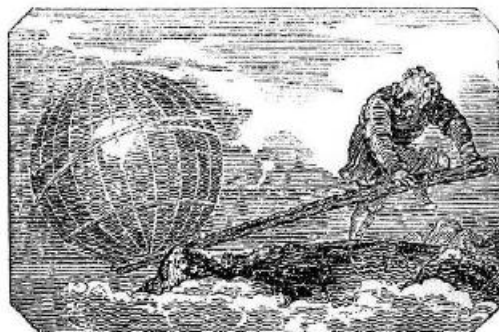
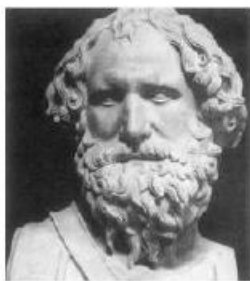


"If every tool, when ordered, or even of its own accord, could do the work that befits it... then there would be no need either of apprentices for the master workers or of slaves for the lords."

Aristotle (384-322 B.C.): Six basic machine elements and description of a robot



Zhang Heng (100 A.D.): South-pointing Chariot (non-magnetic differential mechanism)



Archimedes (287 - 212 B.C.): Using six machine elements for machine design



Al-Jazari (1200 A.D.): Automata and first use of crank to develop the elephant clock

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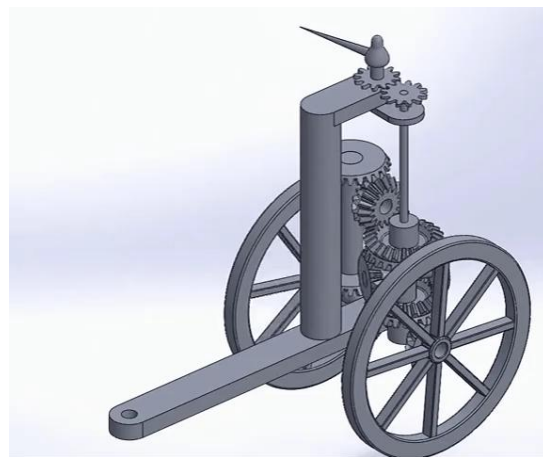
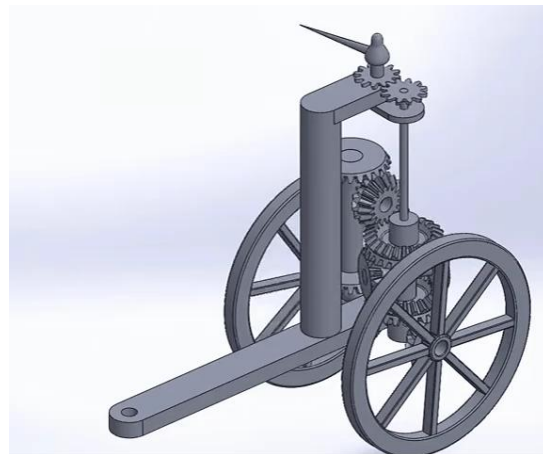
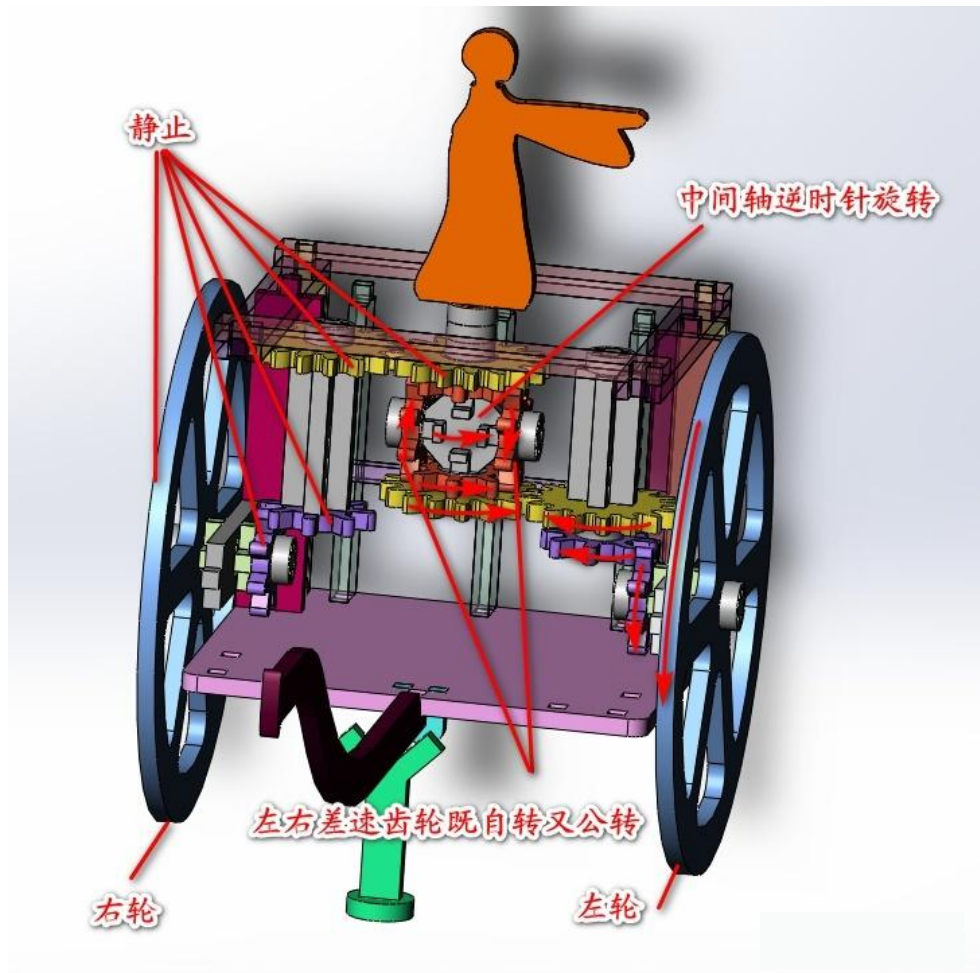
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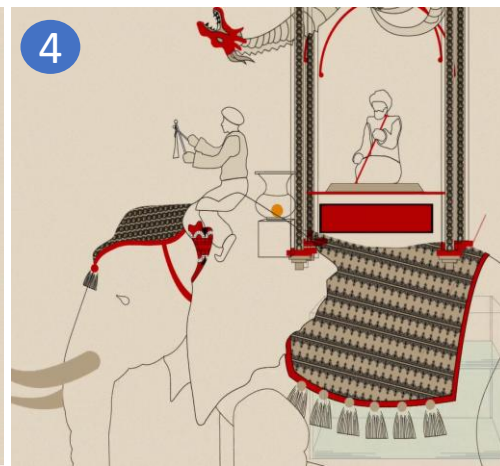
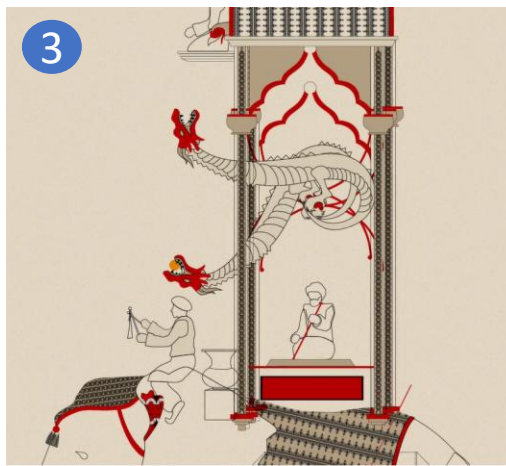
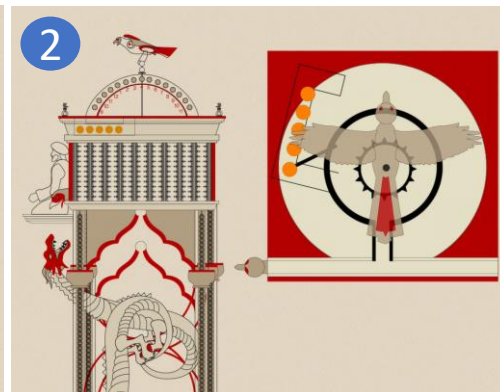
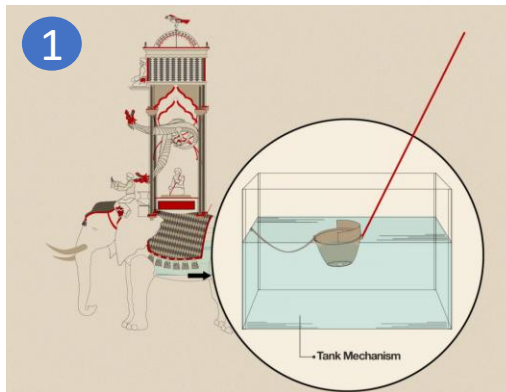
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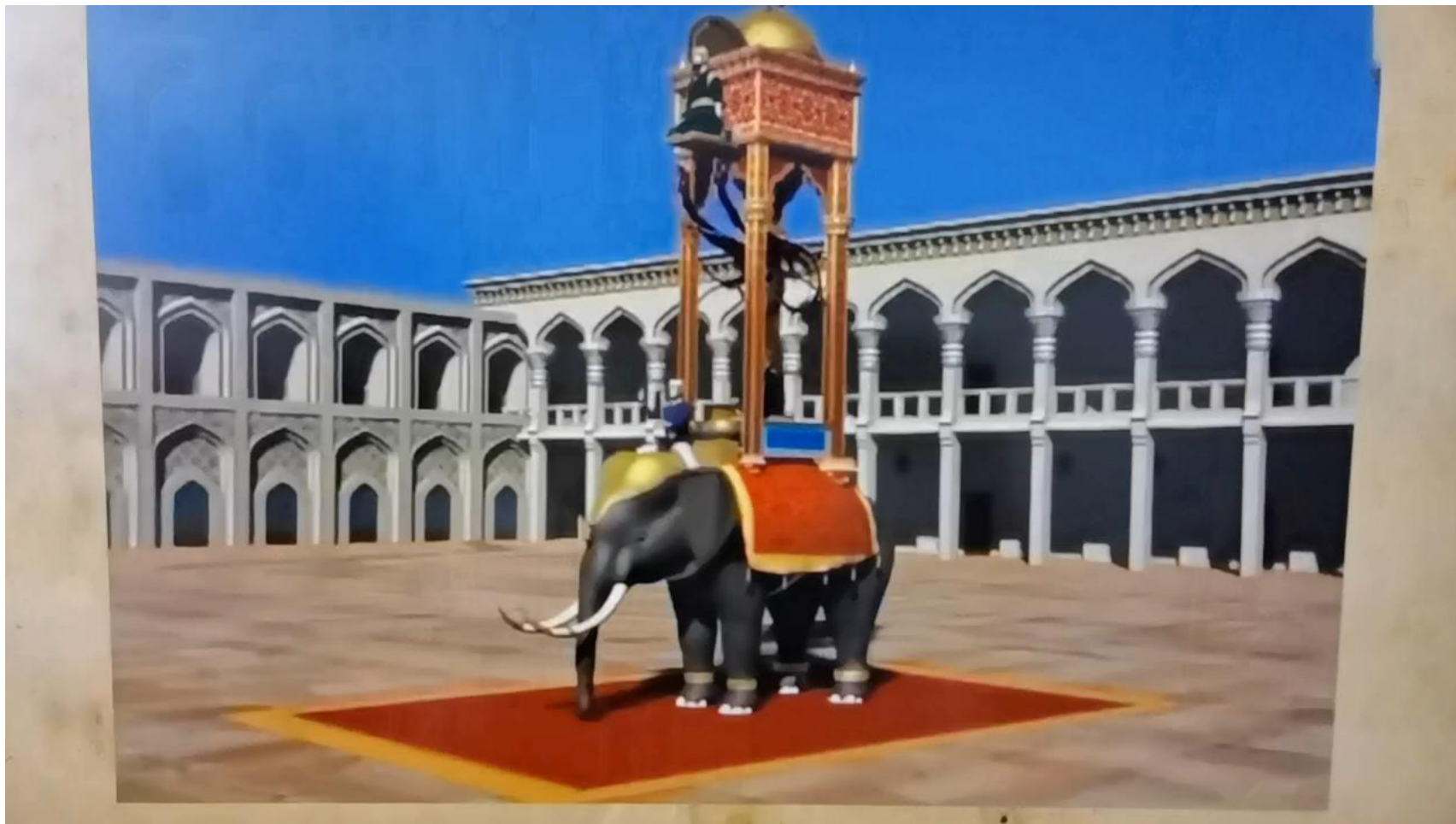
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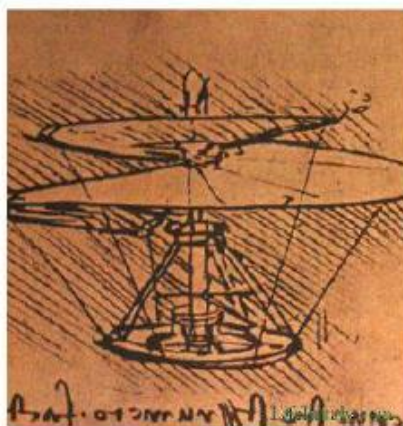
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- Early History (1495 - 1937)

1495 AD

- 1495年左右，达芬奇（Leonardo da Vinci）绘制了拟人机器人的设计



Leonardo da Vinci (1452-1519): Numerous machine designs recorded in Codex Atlanticus, Manuscript B and Codex Madrid

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1500 – 1700 AD

- 希腊人的理解仅限于**简单机械的静力学**。例如：亚历山大的赫隆（公元10-75年）在他的作品“力学”中列出了五种可以“设定运动负荷”的机制；杠杆、轮轴、滑轮、楔块和螺钉，并描述了它们的制造和使用。
- 在文艺复兴时期，简单机械被称呼为机械力的动态，人们从它们可**提升负载距离多远**的角度开始研究。例如：1586年弗兰德工程师西蒙·斯蒂文（Simon Stevin）得出了**斜面的机械优势**，并将其与其它简单机器一并使用。
- 意大利科学家伽利略于1600年在 Le Meccaniche (On Mechanics) 中完成了简单机械的完整动力学理论，展示了**机器作为施力放大器的潜在数学相似性**。他第一个解释了简单机械并不会产生能量，只能改变它。

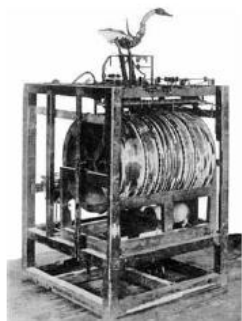
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1700 – 1937 AD

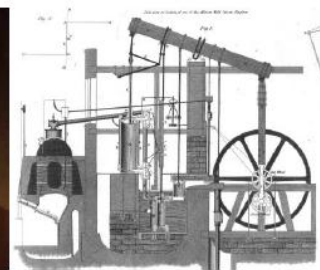
- 在1700年至1900年之间，以欧洲大陆为主体创造了许多与实物大小一样的自动机器，包括由雅克·德·沃坎森（Jacques de Vaucanson）制造的著名机械鸭，它可以吊起脖子，拍打翅膀甚至吞咽食物。
- 瓦特蒸汽机的发明带动了一系列机械机构的创新：行星齿轮、平行运动连杆等



J. Vaucanson (French 1738): Automata and the duck



P. Jaquet-Droz (1770): The writer and piano player



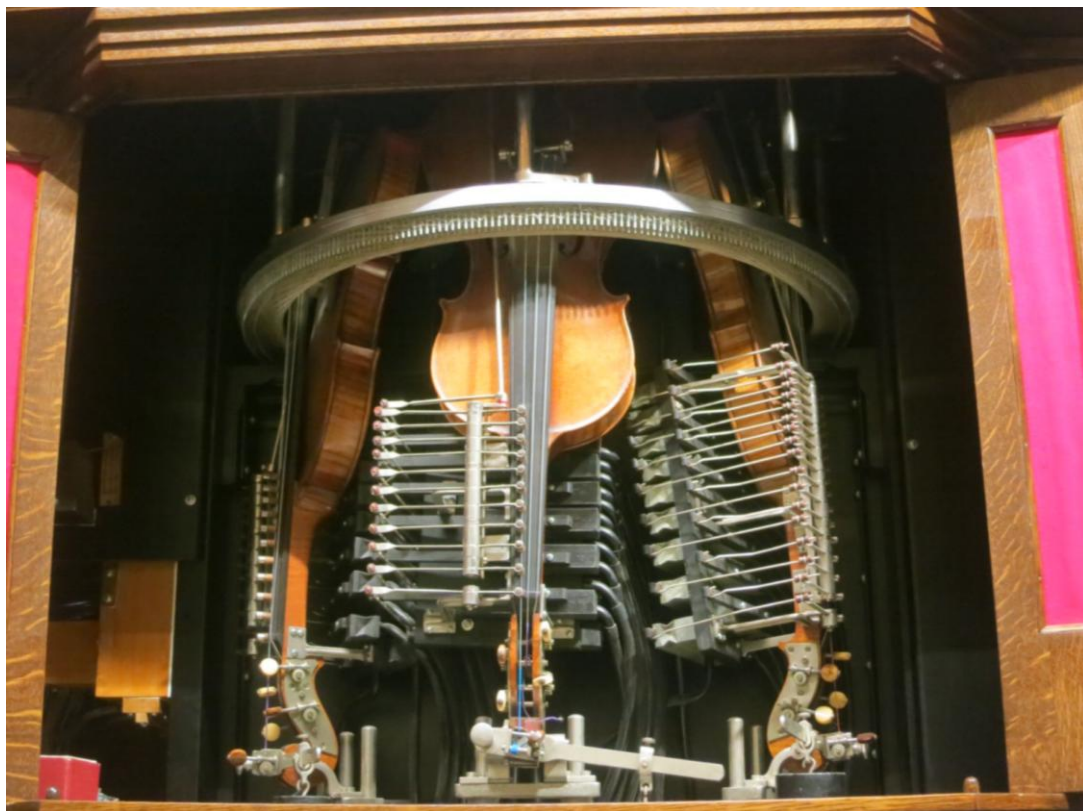
J. Watt(1736-1819): Sun and planet gear, centrifugal governor, parallel motion linkage, and double acting engine.

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乌德勒支音乐盒博物馆 (Museum Speelklok)



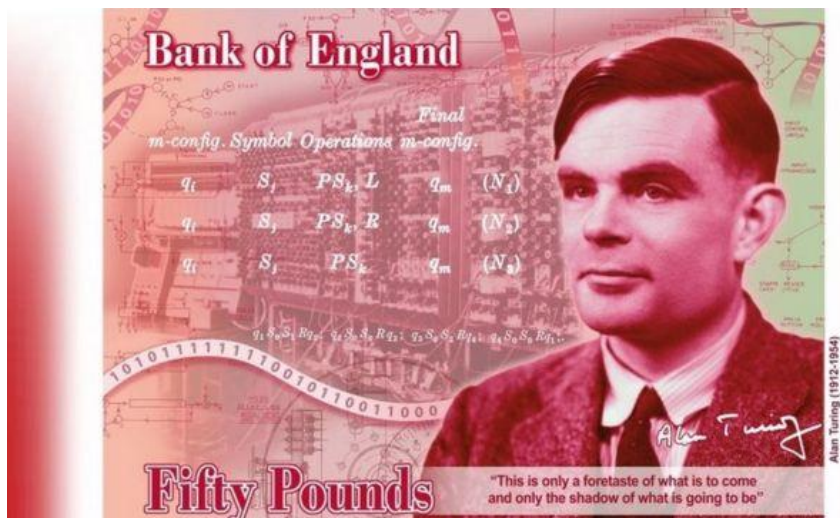
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1937 AD

- 1937年，阿兰·图灵（Alan Turing）发表论文 “On Computable Numbers” 开启了计算机革命。



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1941 AD

1941年，传奇科幻小说家艾萨克·阿西莫夫（Isaac Asimov）在短篇小说“Liar”中描述了机器人技术。阿西莫夫的机器人三定律：

- A robot may not injure a human being or, through inaction, allow a human being to come to harm.
- A robot must obey any orders given to it by human beings, except where such orders would conflict with the First Law.
- A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.



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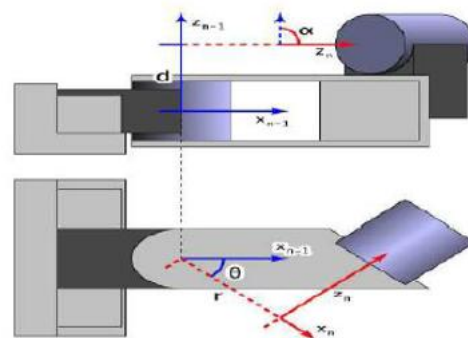
- Modern History (1937 -)

1954 AD

- George Devol和Jo Engelberger设计了第一个可编程机器人手臂，成为第一个工业机器人 **Unimate**，完成危险且重复的任务，列装通用汽车公司的装配线上



George C. Devol filed first robot patent (1954)



J. Denavit and R.S. Hartenberg (1956): Homogeneous transformations for Lower-pair mechanisms

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George Devol and Joseph Engelberger founded Unimation (1961), which installed the first industry robot at a GM plant in Trenton, New Jersey.



American Machine Foundry (AMF 1960) markets Versatran, a cylindrical robot



Stewart and Gough (1960): Stewart platform

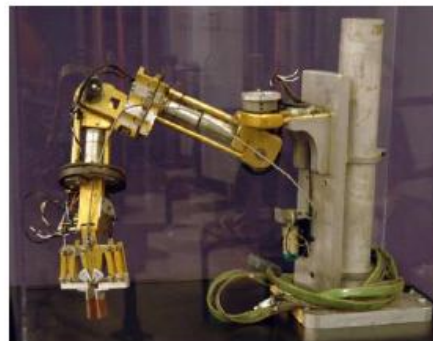
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H. A. Ernst (MIT 1961): Computer control of mechanical arms using touch sensor



Stanford University(1963): Rancho Arm, the first artificial robotic arm to be controlled by a computer



R. Moshier at General Electric (1968): quadrupled walking machine (11 ft tall, 3000lb)



Kawasaki robots in Japan with a patent from Unimation (1968)

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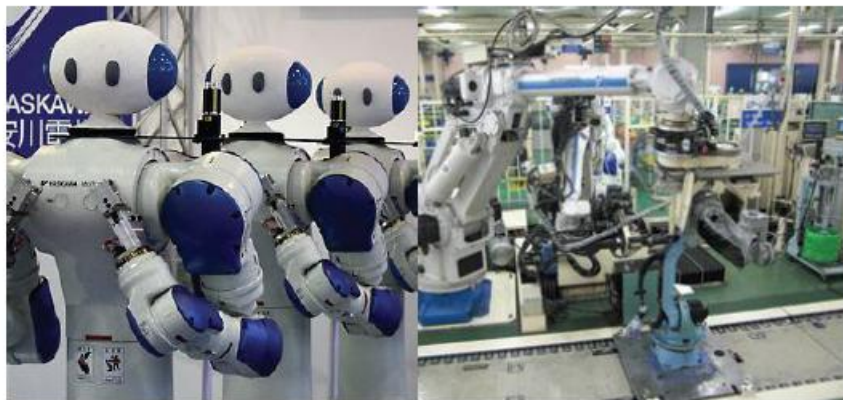
- Modern History (1937 -)



V. Scheinman (1969): The Stanford arm



Draper Lab (1970) (RCC Device), SCARA robots by H. Makino, Japan (1978), Adept Robotics (1982)



Yaskawa engineers coined the term "Mechatronics" (1971)



Waseda University develops Wabot-1 (1973) and Wabot-2 (1980)

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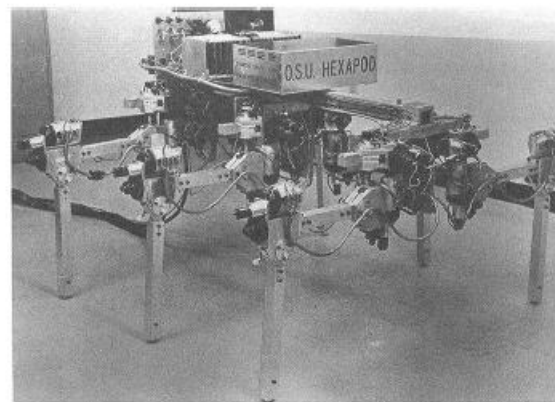
Cincinnati Milacron (1974): (T3 Robots) Payload (100lb)



S. Hirose (1976): The soft gripper



Viking 1 and 2 space probes, equipped with robot arms (1976)



OSU Hexapod (1977)

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1977

发行了第一部《星球大战》电影。乔治·卢卡斯的电影启发了新一代的研究人员探索与机器人共享的人类未来，产生了著名的R2-D2和C-3PO



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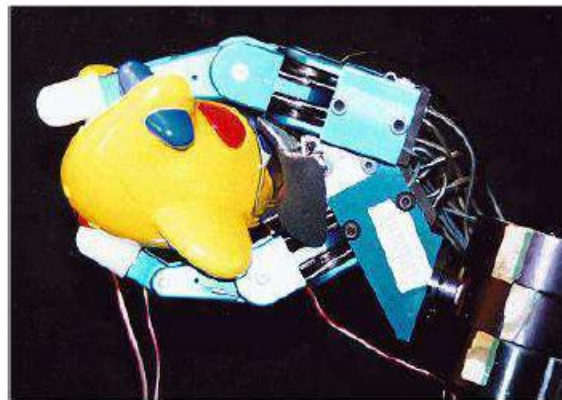
- Modern History (1937 -)



M. Raibert (1980) (RI, CMU & AI lab, MIT): Hopping, Robots, Monoped, biped and Quadpeds. Dynamically stable quadruped robot BigDog created by Boston Dynamics (founded by M. Raibert in 1992) with the NASA Jet Propulsion Laboratory, Caltech (2005)



NASA (1981): Candarm



K. Salisbury (1981): Salisbury Hand

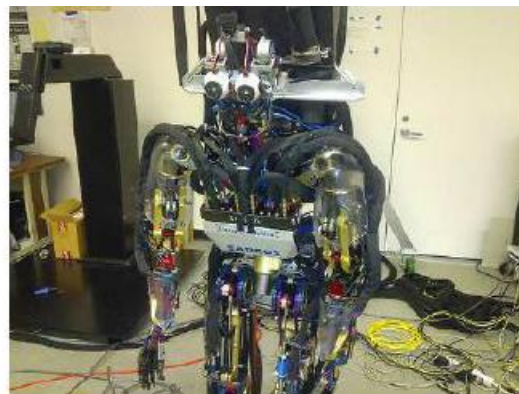
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Fanuc of Japan and General Motors form a joint Venture (1982): Fanuc Robotics America



Sarcos, Utah (1983): Entertainment robot



Odetics Walking robots (1988)



Utah/MIT (1989): Utah/MIT hand

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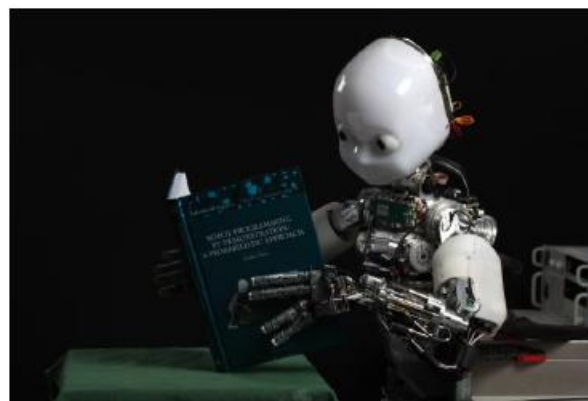
R. Clavel (1991): Delta robot



Da Vinci robot by Intuitive surgical (1995)



Honda (2000): Humanoid Robot, ASIMO



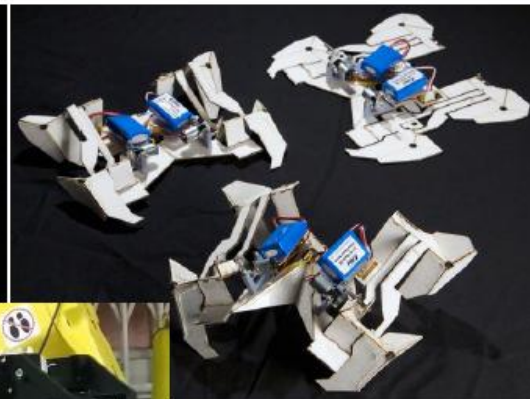
IIT (2004): iCub



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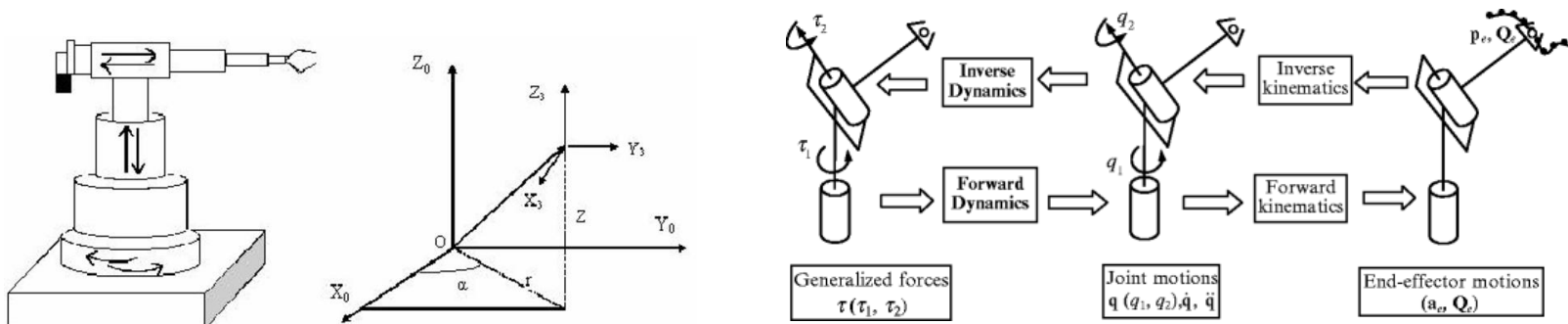
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1.1. 机器人机构学的发展历史与研究的内容

(1) 机构基础、构型原理、不同机器人机构的特点与应用 (模块1, Week 1-7, 21)

(2) 分析机器人机构的数学工具、运动学、动力学、性能评价 (模块2, Week 8-13, 18)

- 与机构学和机器人学联系紧密的数学工具中，人们比较熟悉**线性代数与矩阵理论**，**旋量理论**和**李群、李代数理论**在现代物理学和刚体运动领域取得了成功的应用，也日渐成为现代机构学和机器人学研究的有效分析工具，越来越受到重视。
- 根据**运动学与动力学性能评价设计机构、控制机器人**，**性能评价指标**应具有明确的物理意义，可以用数学方程来描述，具有可计算性，最好可以用一个数字来表示大小：如工作空间 (workspace)、奇异性 (singularity)、速度 (velocity)、承载能力 (payload capability)、刚度 (stiffness)、精度 (accuracy) 等。



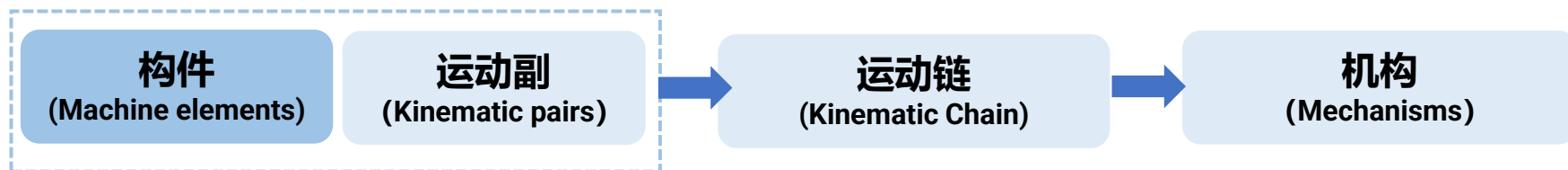
1.2. 机构的基本元素

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机构的基本元素：构件、运动副、运动链

➤ 构件 (Machine elements)

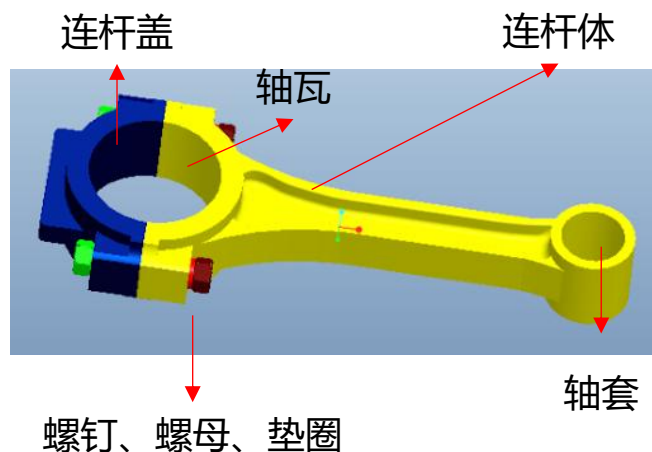


构件定义：机器中每一个**独立的运动单元体**称为一个构件，它可以由单个或数个**机械零件**构成，两个以上的构件可进一步**组成运动副**

零件----独立制造单元体

构件----机器中一个独立运动的单元体

构件可能是一个独立运动的零件，也可以是由若干零件刚性连接在一起的独立运动的整体

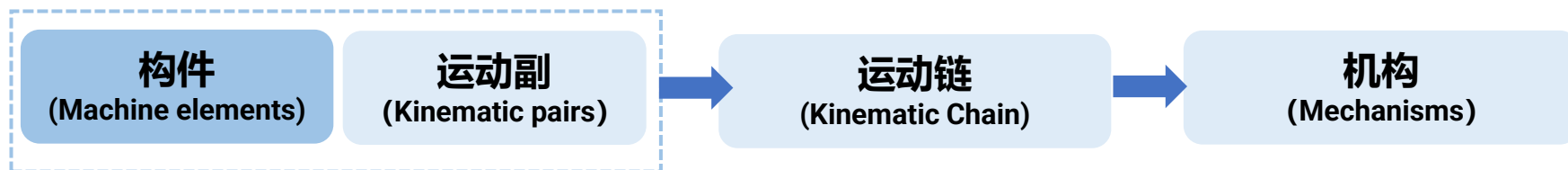


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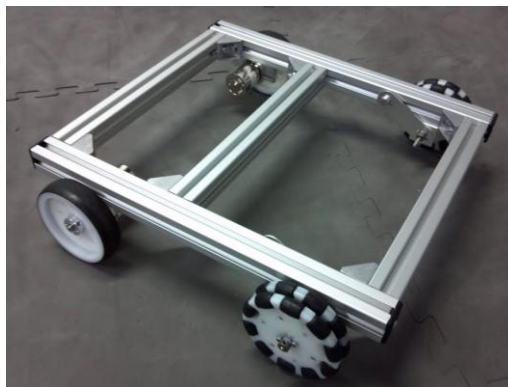
机构的基本元素：构件、运动副、运动链

➤ 构件 (Machine elements)



结构元素：

- 机架(Frames)
- 轴 (Shafts)
- 联轴器 (Shaft couplings)
- 轴承类 (Bearings)
- 紧固件 (Fasteners)
- 键(Keys)、花键(Splines)、开口销(Cotter Pins)
- 密封件(Seals)

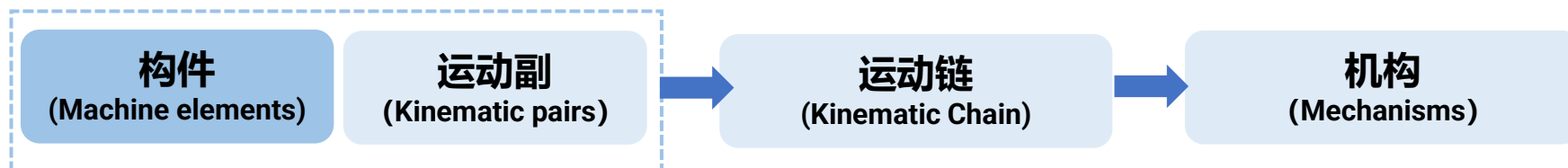


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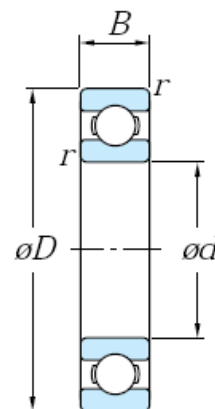
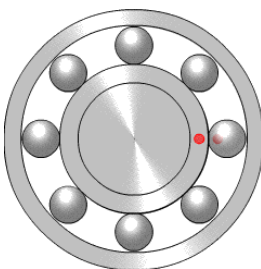
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- 密封件(Seals)



Open



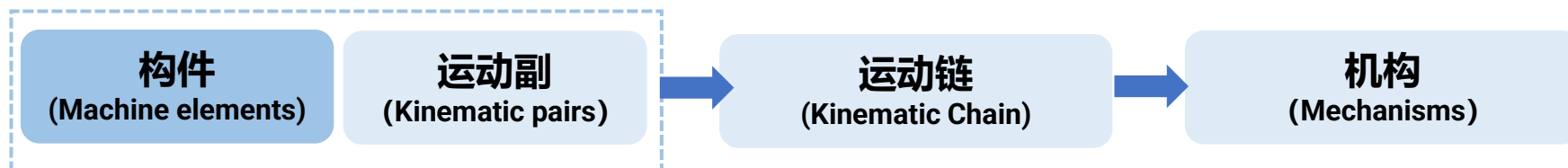
Rubber & Metal Sealed

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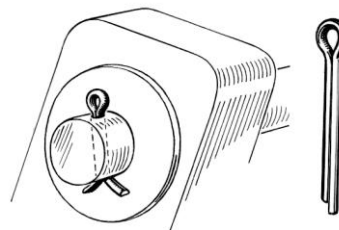
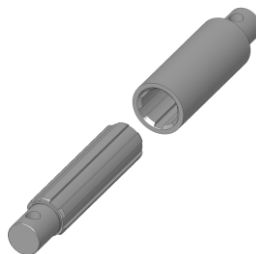
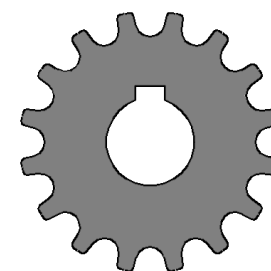
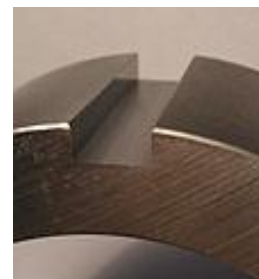
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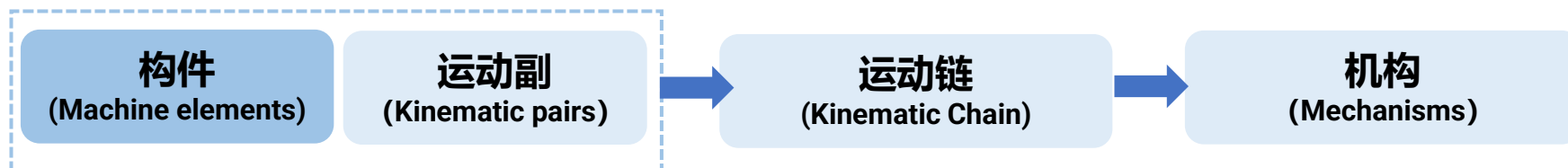


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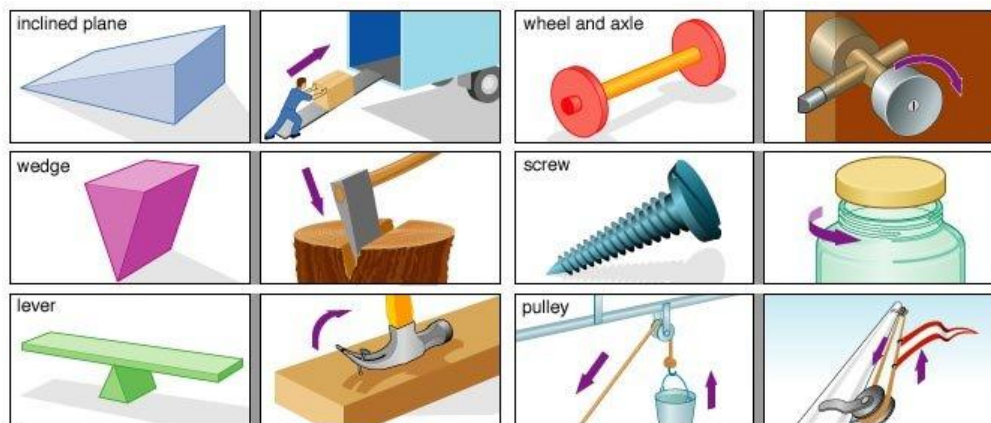
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➤ 构件 (Machine elements)



机械元件：

- 简单机械 (Simple machine)
- 齿轮/齿系(Gear/Gear train)
- 凸轮和从动(Cam and follower)
- 皮带(Belt)、绳索(Cable/Chain)
- 连杆(Linkage)



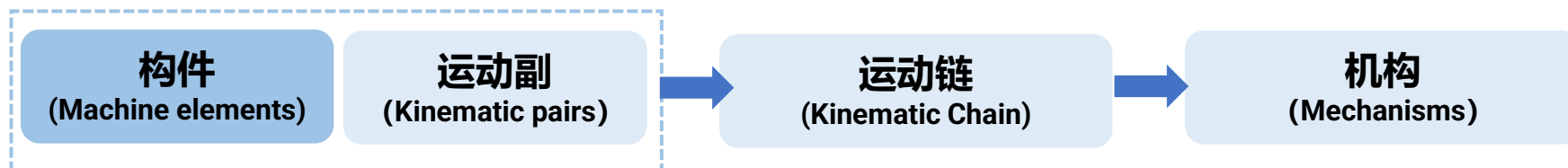
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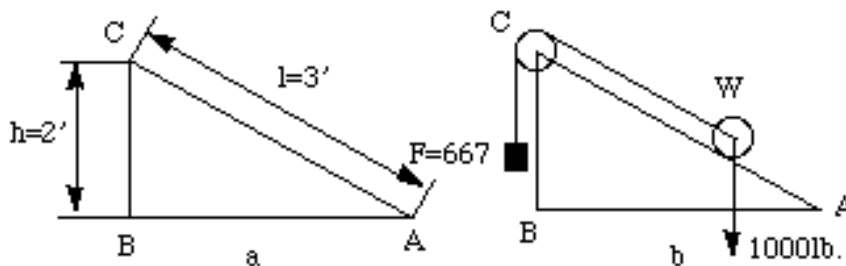
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机械元件：

- 简单机械 (Simple machine)
- 齿轮/齿系(Gear/Gear train)
- 凸轮和从动(Cam and follower)
- 皮带(Belt)、绳索(Cable/Chain)
- 连杆(Linkage)

□ 斜面(Inclined Plane) & 楔(wedge) & 螺杆(screw)



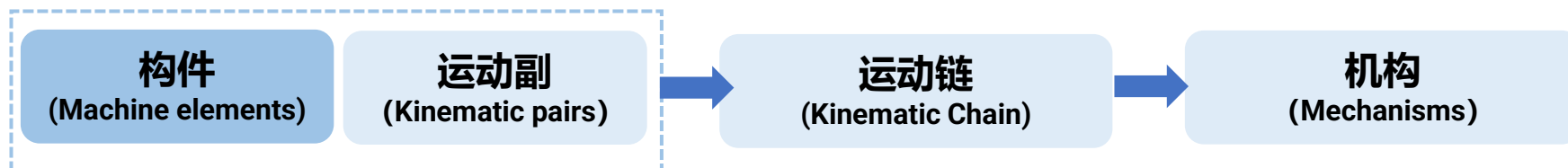
$$\frac{F}{W} = \frac{h}{l}$$

1. 机器人与机构学基础

1.2. 机构的基本元素

机构的基本元素：构件、运动副、运动链

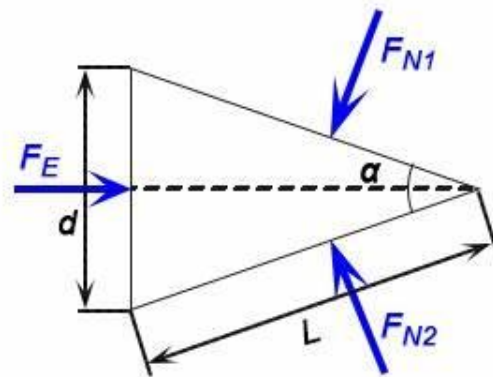
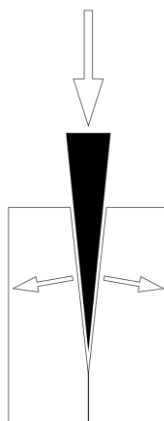
➤ 构件 (Machine elements)



机械元件：

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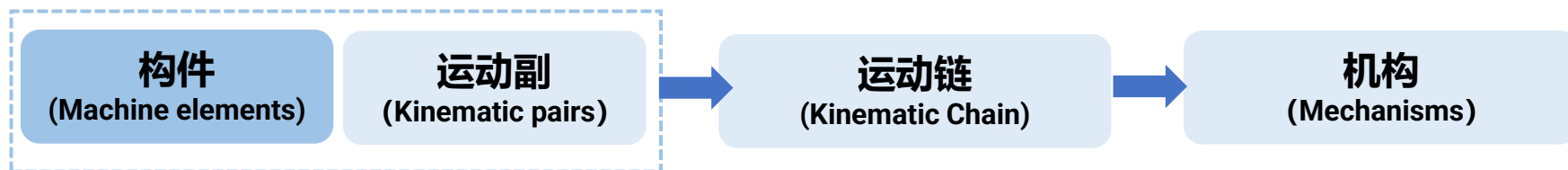


1. 机器人与机构学基础

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机构的基本元素：构件、运动副、运动链

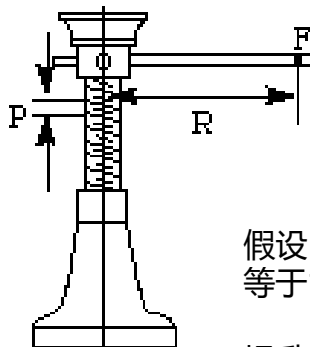
► 构件 (Machine elements)



机械元件：

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- 连杆(Linkage)

□ 斜面(Inclined Plane) & 楔(wedge) & 螺杆(screw)



$$F \times 2\pi R = W \times P$$

$$F = \frac{W \times P}{2\pi R}$$

假设 R 等于18 cm, P 等于1/8 cm, 要提升的重量等于1500 kg, 那么 F 处所需的力为1.6579 kg。

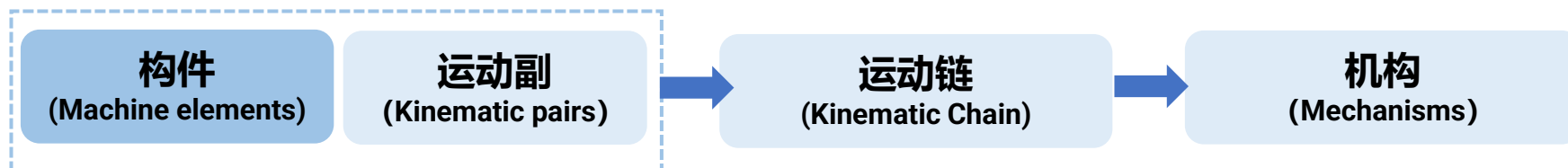
提升的重量的移动速度比 F 处施加的力慢得多。

1. 机器人与机构学基础

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机构的基本元素：构件、运动副、运动链

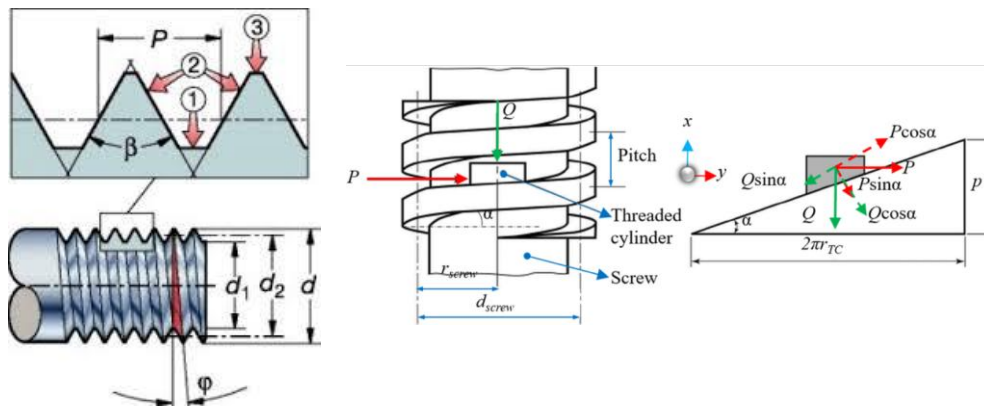
➤ 构件 (Machine elements)



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□ 斜面(Inclined Plane) & 楔(wedge) & 螺杆(screw)

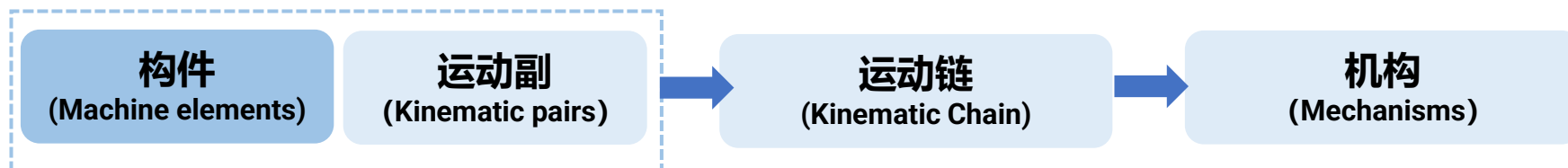


1. 机器人与机构学基础

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机构的基本元素：构件、运动副、运动链

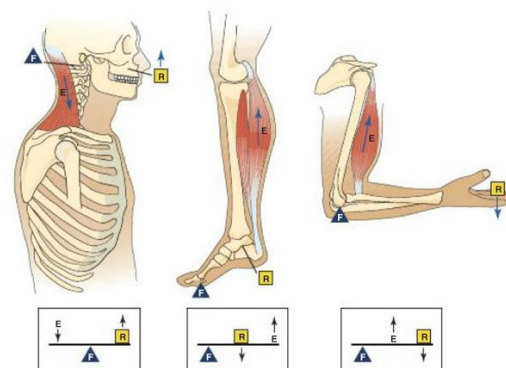
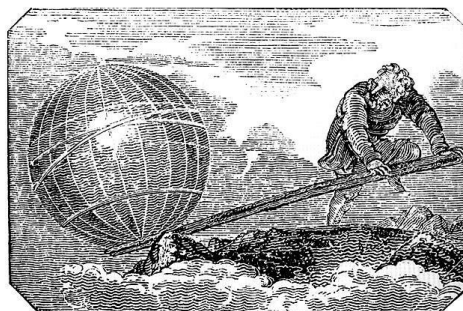
► 构件 (Machine elements)



机械元件：

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- 连杆(Linkage)

□ 杠杆 (lever)

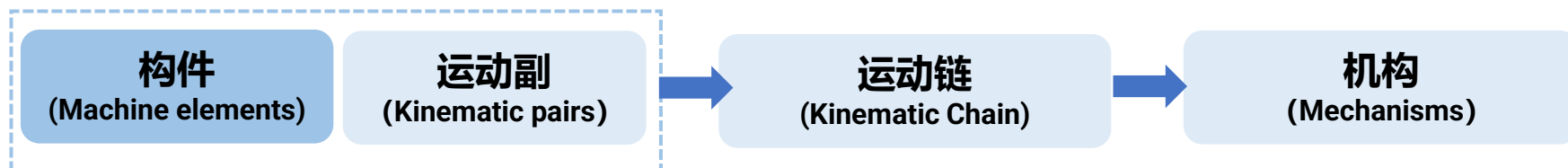


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机构的基本元素：构件、运动副、运动链

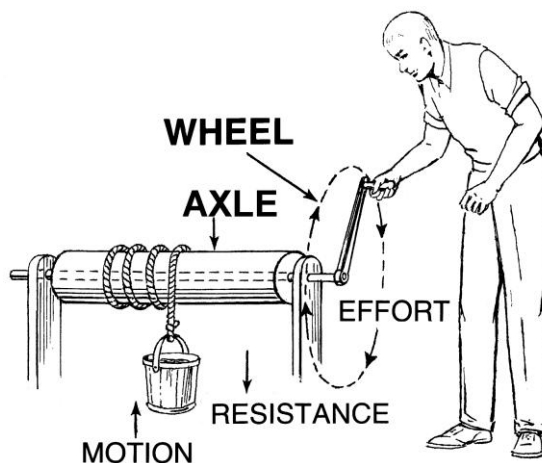
➤ 构件 (Machine elements)



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- 连杆(Linkage)

□ 轮轴 (wheel & axle)

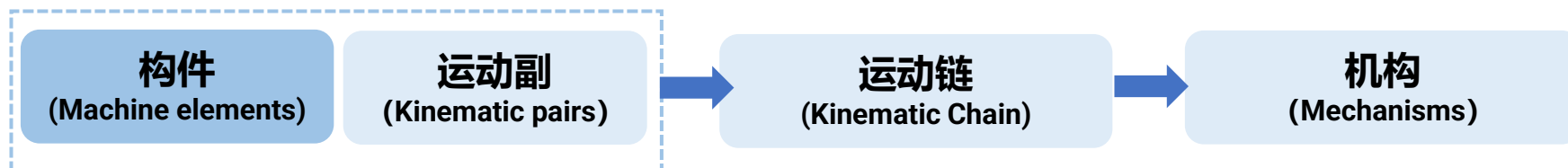


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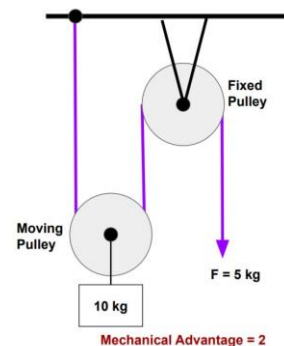
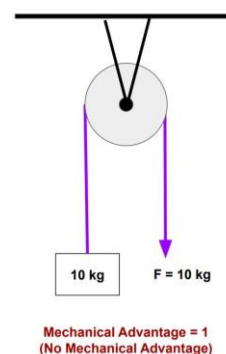
➤ 构件 (Machine elements)



机械元件：

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- 皮带(Belt)、绳索(Cable/Chain)
- 连杆(Linkage)

□ 滑轮 (pulley)

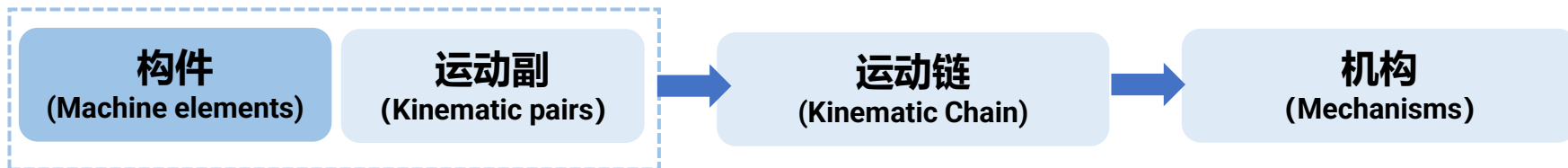


1. 机器人与机构学基础

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机构的基本元素：构件、运动副、运动链

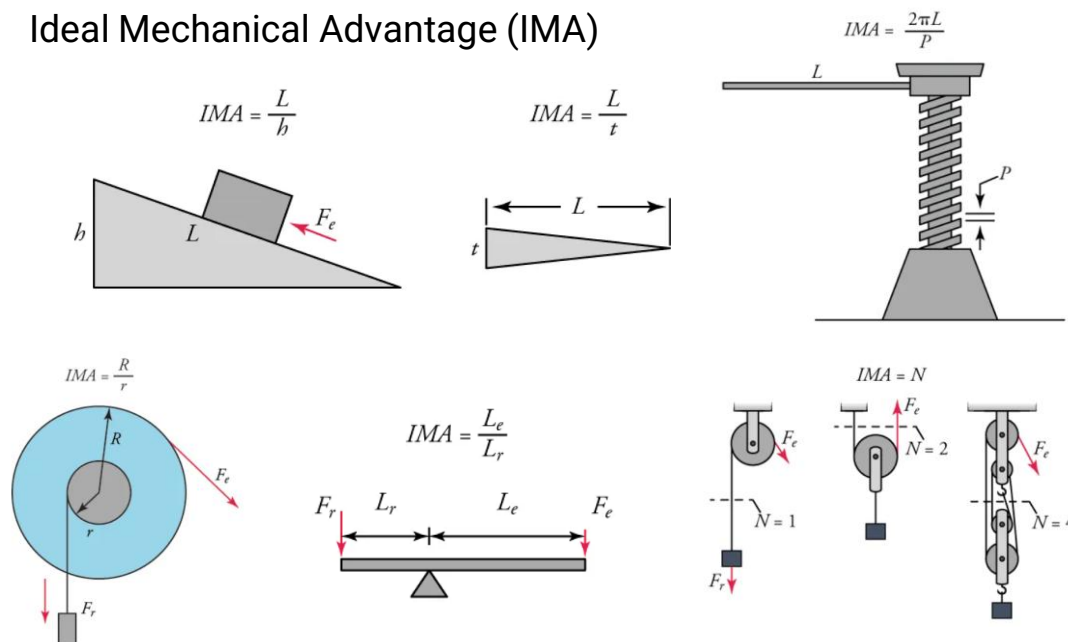
➤ 构件 (Machine elements)



Ideal Mechanical Advantage (IMA)

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- 连杆(Linkage)

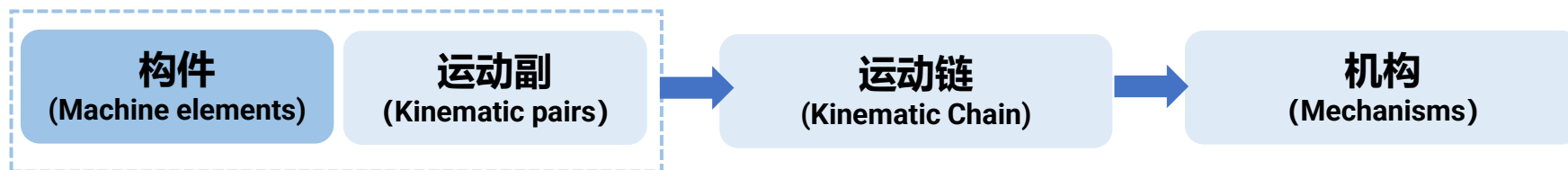


1. 机器人与机构学基础

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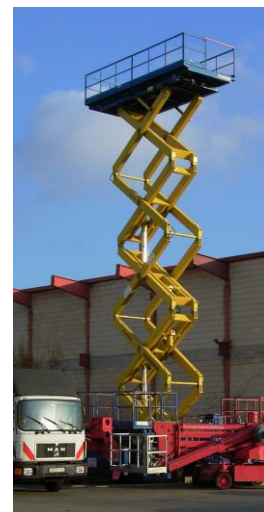
机构的基本元素：构件、运动副、运动链

➤ 构件 (Machine elements)



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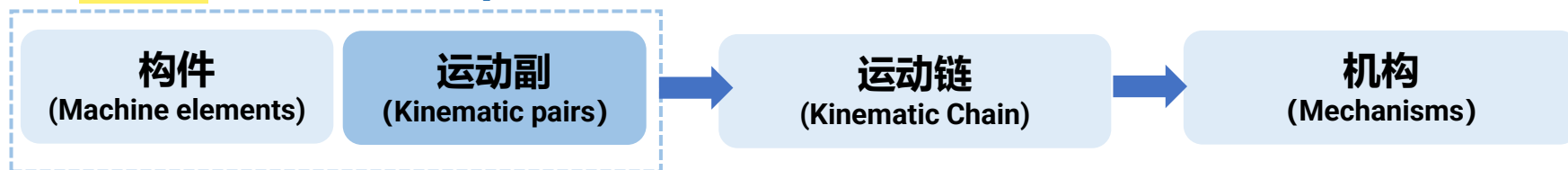


1. 机器人与机构学基础

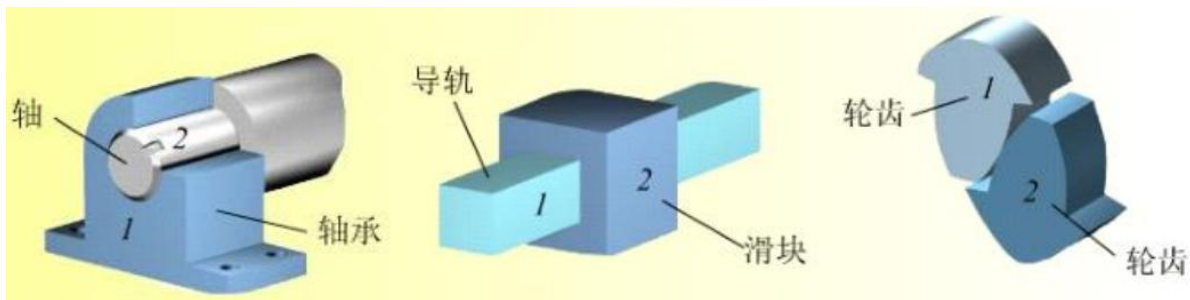
1.2. 机构的基本元素

机构的基本元素：构件、运动副、运动链

➤ 运动副 (Kinematic pairs)



- 两个构件之间接触式的可动联接，运动副作用是限制构件间的**相对运动 (Relative motion)**，或者说使构件之间产生受约束的相对运动
- **自由度 (Degrees of freedom)** 的数目减少，这种限制称为**约束 (Constrain)**



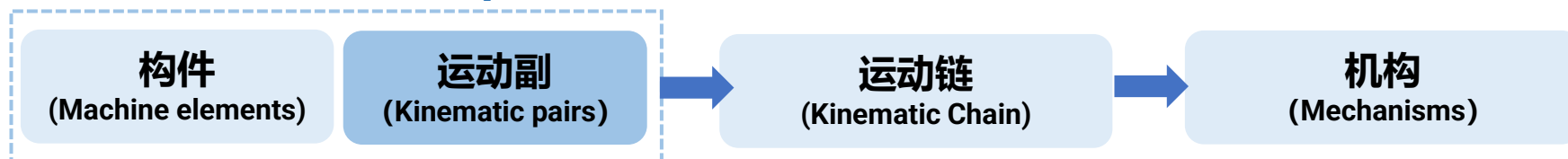
- a) 两个构件
 - b) 直接接触
 - c) 有相对运动
- 三个条件，缺一不可**

1. 机器人与机构学基础

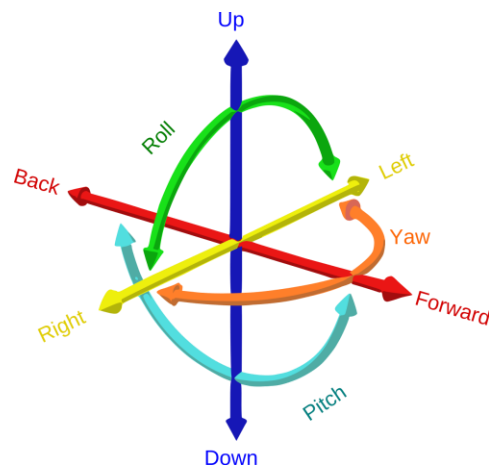
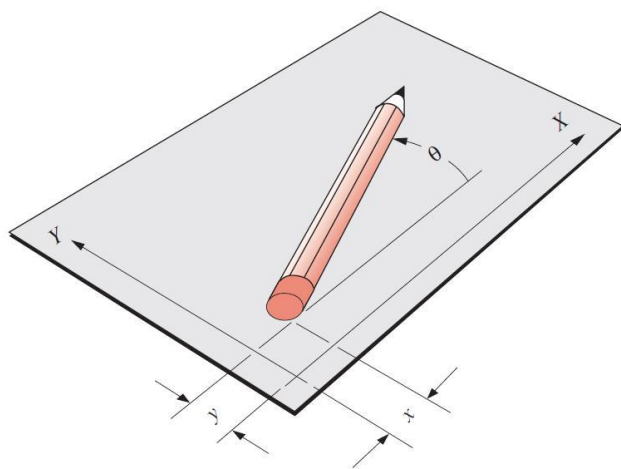
1.2. 机构的基本元素

机构的基本元素：构件、运动副、运动链

➤ 运动副 (Kinematic pairs)



自由度：机构具有确定运动时所必须给定的独立运动参数的数目（亦即使机构的位置得以确定，必须给定的独立的广义坐标的数目），其数目常以F表示。

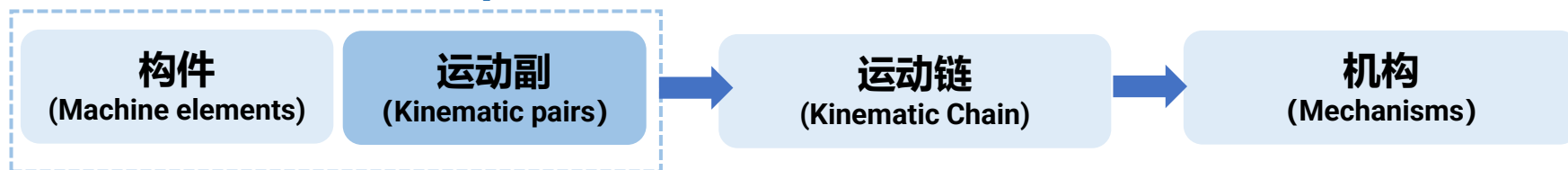


1. 机器人与机构学基础

1.2. 机构的基本元素

机构的基本元素：构件、运动副、运动链

► 运动副 (Kinematic pairs)



1) 按运动副的接触方式分：高副、低副

凡是以面接触的运动副称为低副，而以点或线相接触的运动副称为高副

2) 按相对运动的范围分：平面运动副、空间运动副

运动构件间只能作平面运动称之为平面运动副，否则称为空间运动副

3) 按运动副引入的约束数分：

引入一个约束数的运动副称为I级副，依此类推II、III、IV、V级副

4) 按运动副接触部分的几何形状分：

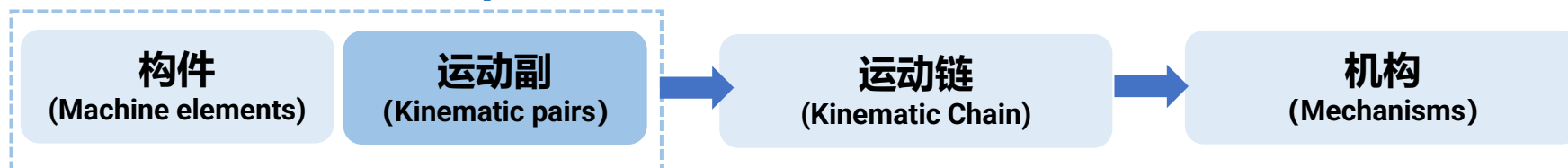
转动副、移动副、螺旋副、圆柱副、平面副、球面副、点接触高副、线接触高副

1. 机器人与机构学基础

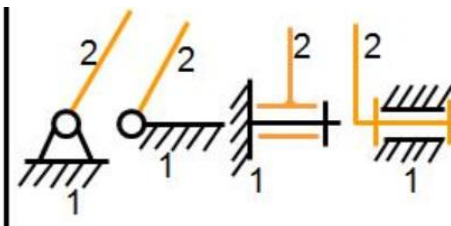
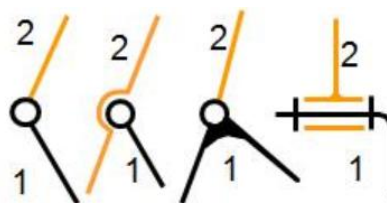
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机构的基本元素：构件、运动副、运动链

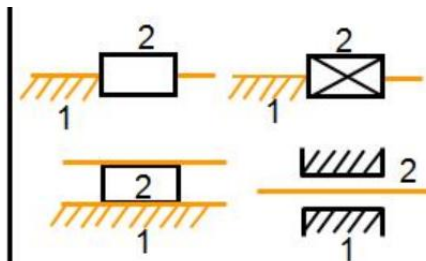
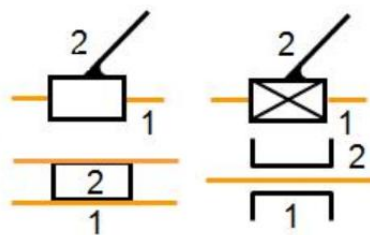
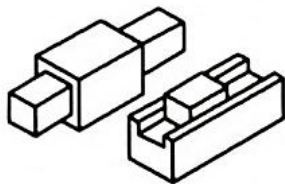
➤ 运动副 (Kinematic pairs)



转动副 (R)
平面V级低副
自由度: 1R



移动副 (T)
平面V级低副
自由度: 1T

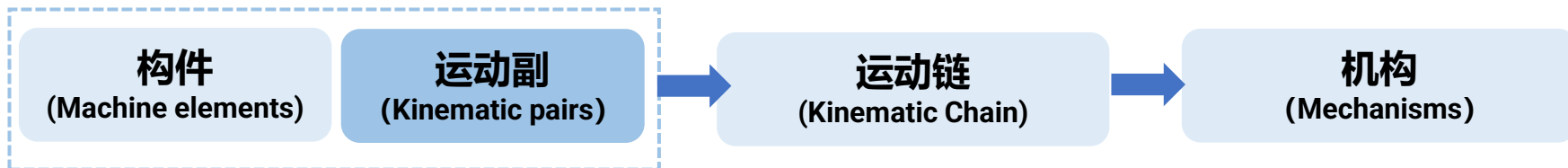


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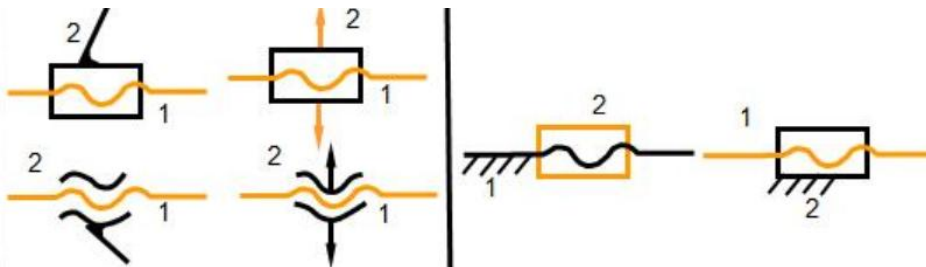
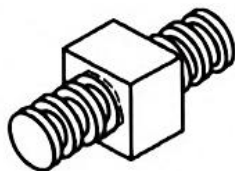
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机构的基本元素：构件、运动副、运动链

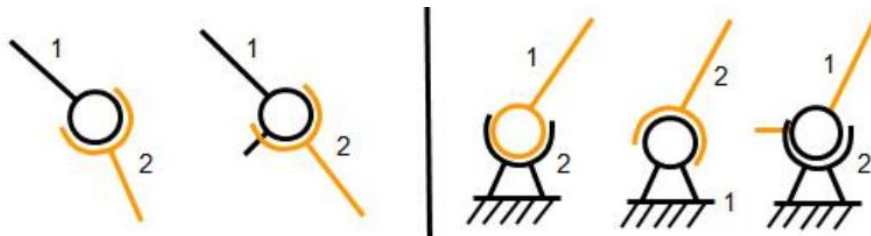
➤ 运动副 (Kinematic pairs)



螺旋副 (H)
空间V级低副
自由度: 1R或1T



球面副 (S)
空间III级低副
自由度: 3R

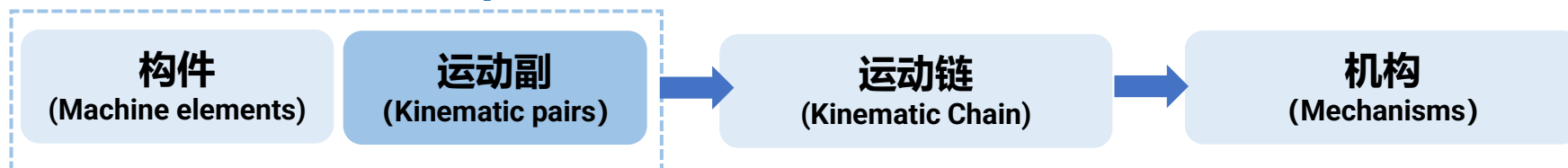


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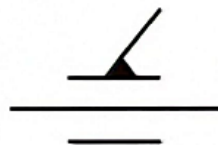
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机构的基本元素：构件、运动副、运动链

► 运动副 (Kinematic pairs)



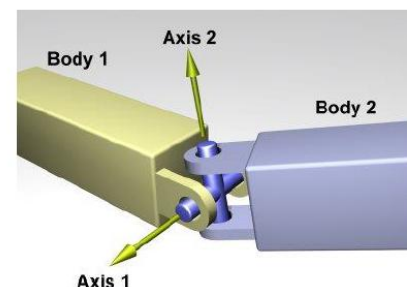
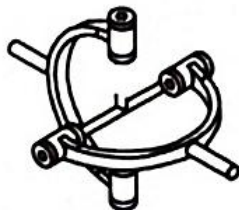
平面副 (E)
平面III级低副
自由度: 1R2T



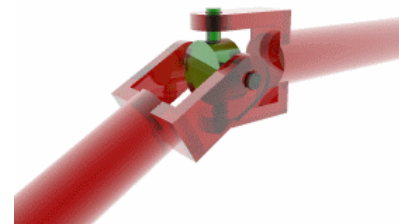
圆柱副 (C)
空间IV级低副
自由度: 1R1T



虎克铰 (U)
空间IV级低副
自由度: 2R



Universal joint

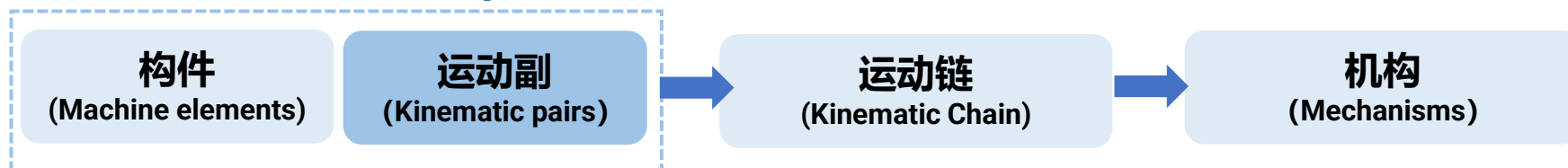


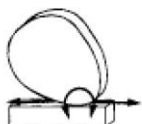
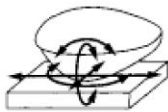
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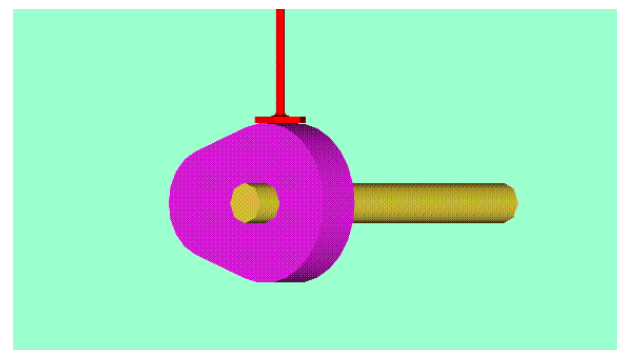
1.2. 机构的基本元素

机构的基本元素：构件、运动副、运动链

➤ 运动副 (Kinematic pairs)



- | | | | |
|---|-----------------------|---|--|
| 1 | Cylindrical roller |  | Roller rotates about this line at this instant in its motion. Roller does not slip on the surface on which it rolls. |
| 2 | Cam pair |  | Cam rolls and slides on follower. |
| 3 | Rolling ball |  | Ball rolls without slipping. |
| 4 | Ball in cylinder |  | Ball can rotate about any axis through its center and slide along cylinder axis. |
| 5 | Spatial point contact |  | Body can rotate about any axis through the contact point and slide in any direction in the tangent plane. |

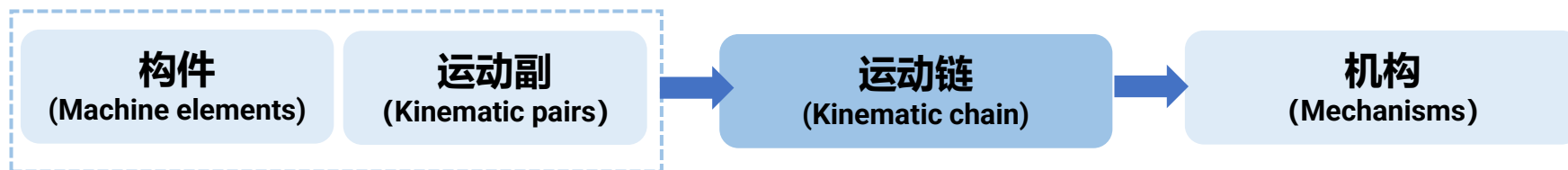


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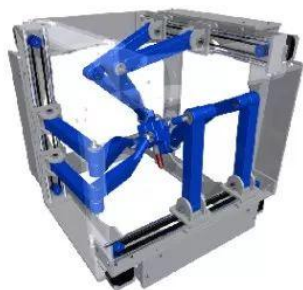
机构的基本元素：构件、运动副、运动链

➤ 运动链 (Kinematic chain)



两个以上的构件通过运动副联接而构成的相对可动的构件系统

- 闭式链：运动链中的各构件构成首末封闭的系统，如并联机器人
- 开式链：运动链中的各构件未构成首末封闭的系统，如串联机器人

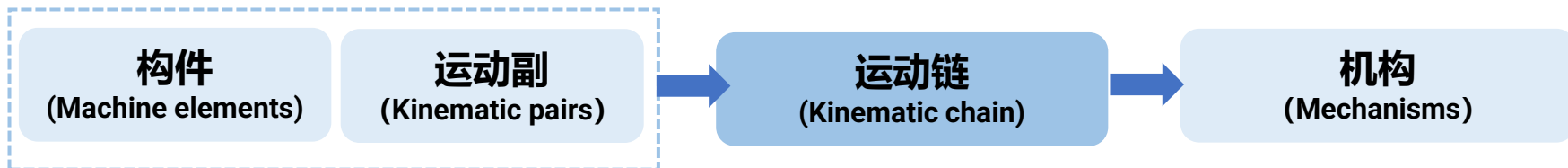


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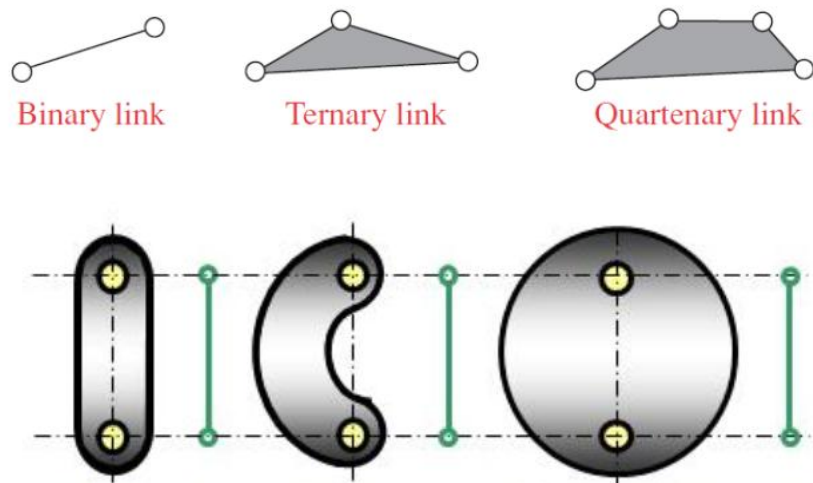
机构的基本元素：构件、运动副、运动链

➤ 运动链 (Kinematic chain)



杆件的表示方法:

杆、轴构件	
固定构件	
同一构件	

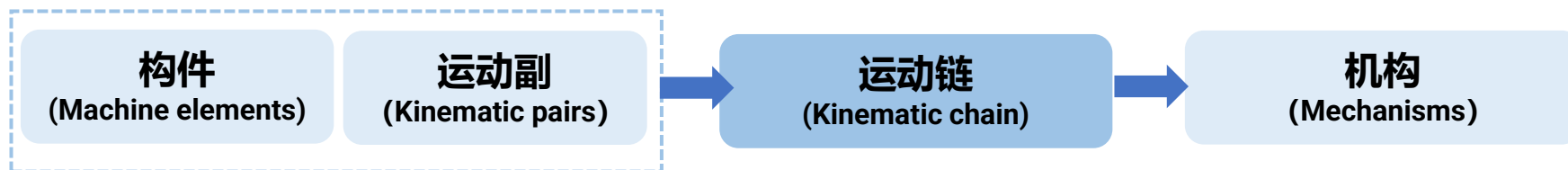


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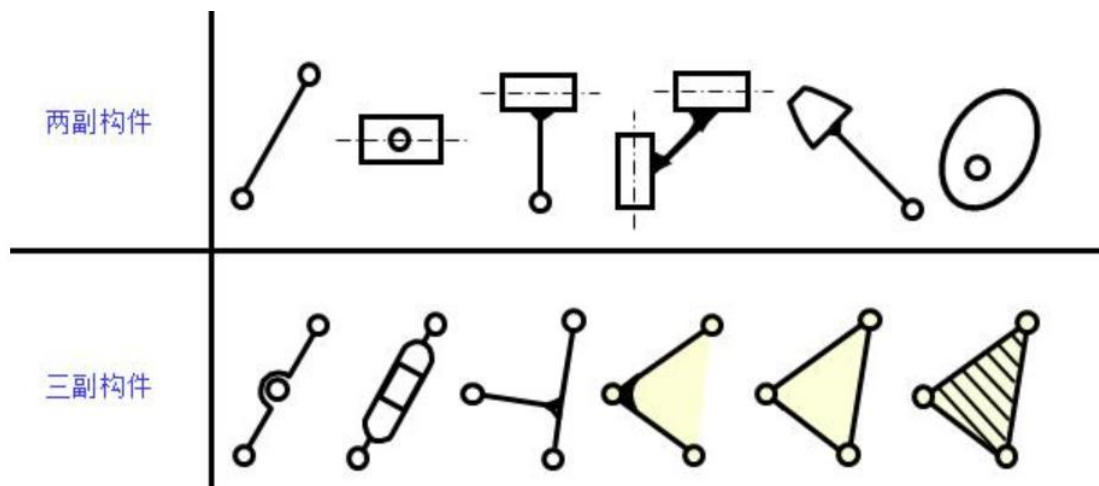
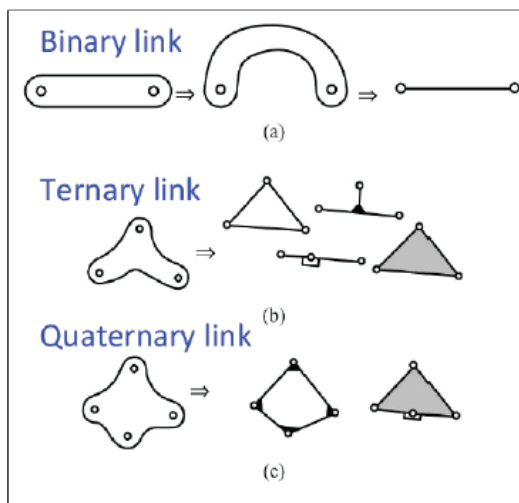
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机构的基本元素：构件、运动副、运动链

➤ 运动链 (Kinematic chain)



杆件的表示方法:

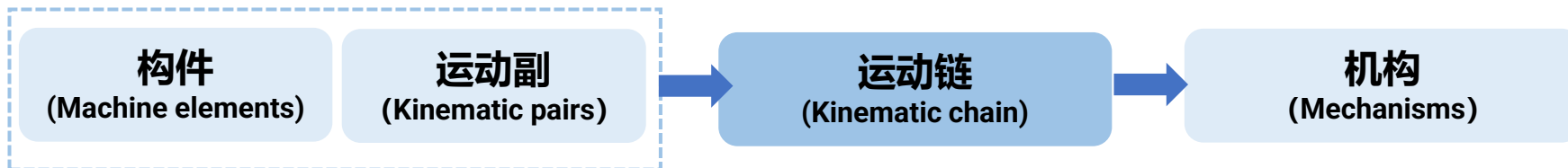


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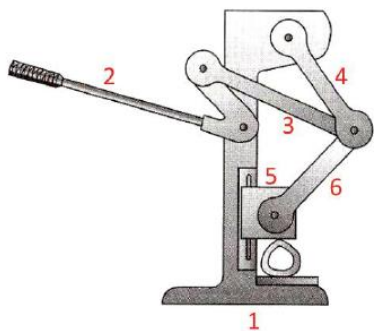
1.2. 机构的基本元素

机构的基本元素：构件、运动副、运动链

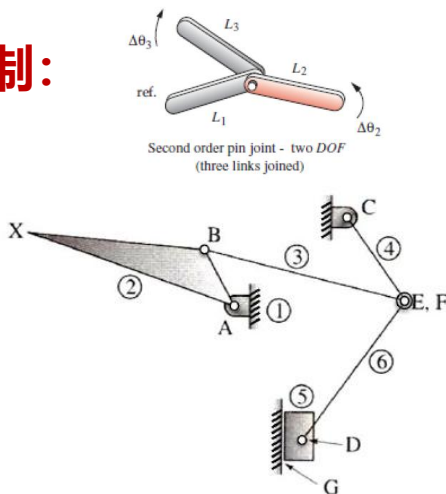
➤ 运动链 (Kinematic chain)



机构运动简图的绘制:



Functional sketch



Kinematic diagram

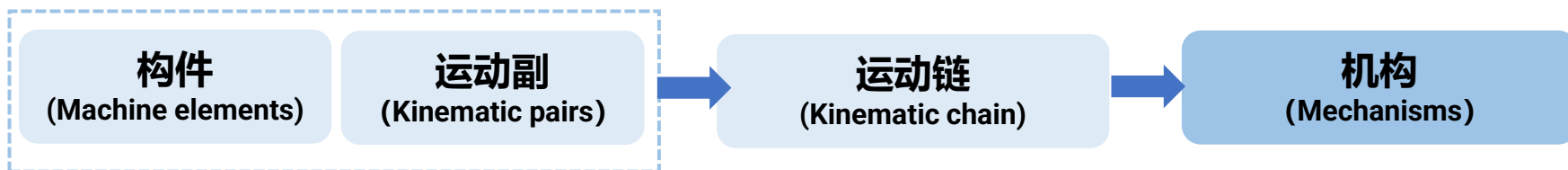
- 分析机械的动作原理、组成情况和运动情况;
- 逐一分析每两个构件间相对运动的性质, 确定运动副的类型和数目;
- 选择适当的机构运动瞬时位置和比例尺, 定出各运动副的相对位置, 并用各运动副的代表符号、常用机构的运动简图符号和简单线条, 绘制机构运动简图;
- 从原动件开始, 按运动传递顺序标出各构件的编号和运动副代号。

1. 机器人与机构学基础

1.2. 机构的基本元素

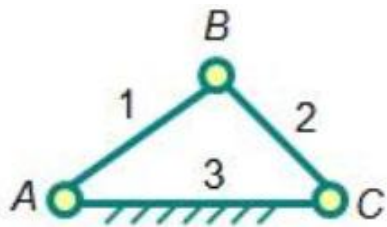
机构的基本元素：构件、运动副、运动链

► 机构 (Mechanisms)



将运动链中的某个构件或几个加以固定，而让另一个或几个构件按给定运动规律相对**固定构件**运动，如果运动链中**其余各活动构件都具有确定的相对运动**，则此运动链称为机构。其中的固定构件称作机架(base)

- 运动链成为机构的首要条件：运动链自由度必须大于零
- 在 $F > 0$ 的条件下，进一步判断是否具有确定的运动（原动件个数和自由度关系）

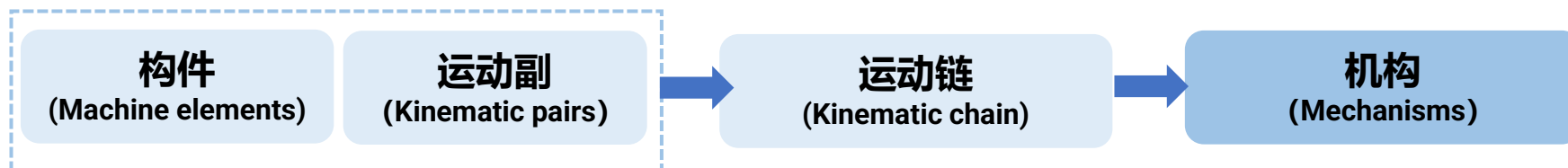


1. 机器人与机构学基础

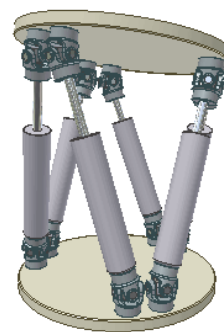
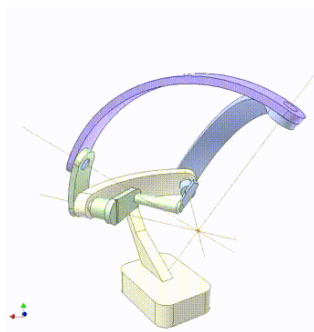
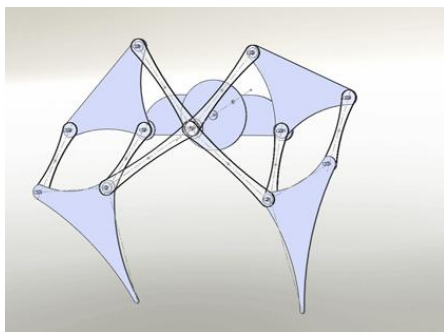
1.2. 机构的基本元素

机构的基本元素：构件、运动副、运动链

► 机构 (Mechanisms)



- **根据机构各构件的相对运动：**平面机构 (Planar mechanism)、球面机构 (Spherical mechanism)、空间机构 (Spatial mechanism)
- **根据构件或运动副的柔度：**刚性机构 (Rigid mechanism)、弹性机构 (Flexible mechanism)、柔性机构 (Compliant mechanism)



1.3. 自由度与约束

1. 机器人与机构学基础

1.3. 自由度与约束

自由度与约束的计算方法

- 运动链的自由度：确定运动链中各构件相对于其中某一构件的位置所需的独立参变量的数目，在平面内作自由运动的构件具有3个独立的相对运动；在三维空间作自由运动的构件具有6个独立的相对运动
- 约束：运动副对构件间相对运动的限制作用，约束个数等于其自由度减少的个数
- 考察由 L 个构件组成的运动链，活动构件数 $n = L - 1$

构件	I级副	II级副约	III级副约	IV级副	V级副约
总自由度	约束数	束数	束数	约束数	束数
$6n$	p_1	$2p_2$	$3p_3$	$4p_4$	$5p_5$

运动链自由度：

$$F = 6n - (p_1 + 2p_2 + 3p_3 + 4p_4 + 5p_5) = 6n - \sum_{k=1}^5 kp_k$$

具有 q 个公共约束的运动链，其自由度计算公式为：

$$F = (6 - q)n - \sum_{k=q+1}^5 (k - q)p_k$$

平面运动链 $q = 3$ ，其自由度计算公式为：

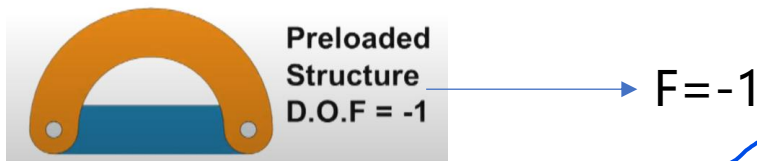
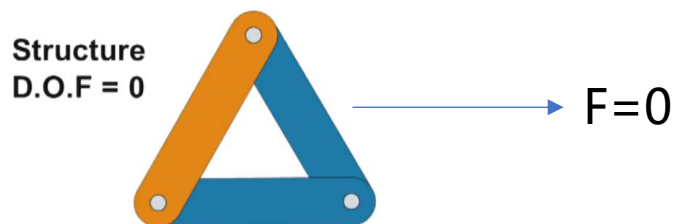
$$F = (6 - 3)n - \sum_{k=4}^5 (k - 3)p_k \Rightarrow F = 3n - 2p_5 - p_4$$

$$\Rightarrow F = 3n - 2p_L - p_H$$

平面机构的自由度计算：

$$F = 3n - 2P_L - P_H$$

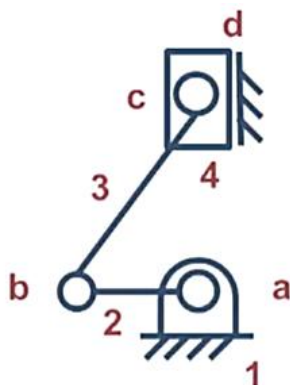
n 为活动构件个数； P_L 为低副个数； P_H 为高副个数



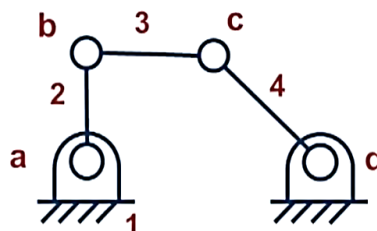
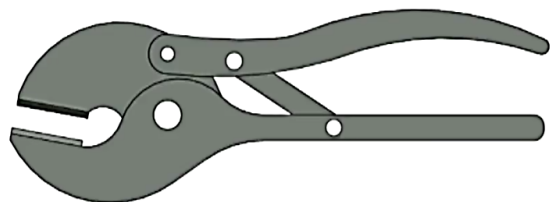
1. 机器人与机构学基础

1.3. 自由度与约束

自由度与约束的计算方法



$$F = 3(l - 1) - 2P_1 - P_2$$
$$F = 3(4 - 1) - 2 \times 4 - 0 = 1$$

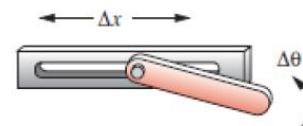
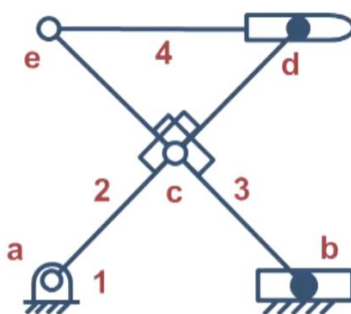
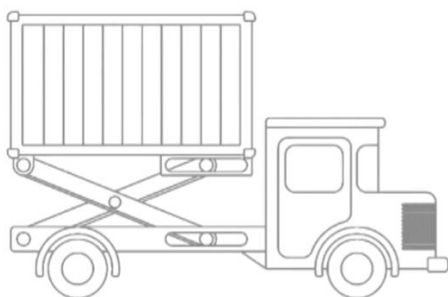


$$F = 3(l - 1) - 2P_1 - P_2$$
$$F = 3(4 - 1) - 2 \times 4 - 0 = 1$$

1. 机器人与机构学基础

1.3. 自由度与约束

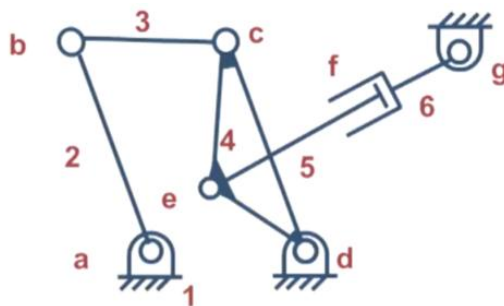
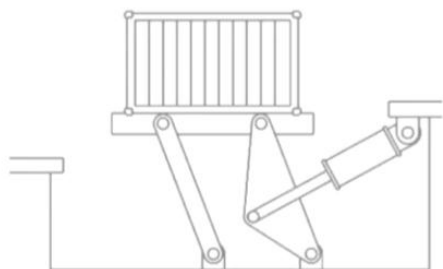
自由度与约束的计算方法



Pin in slot (form closed)

$$F = 3(l - 1) - 2P_1 - P_2$$

$$F = 3(4 - 1) - 2 \times 3 - 2 = 1$$



$$F = 3(l - 1) - 2P_1 - P_2$$

$$F = 3(6 - 1) - 2 \times 7 - 0 = 1$$

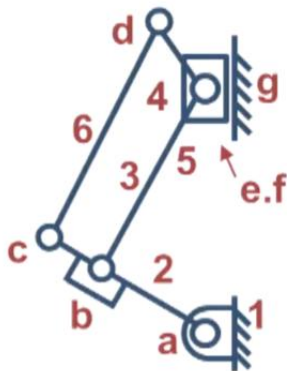
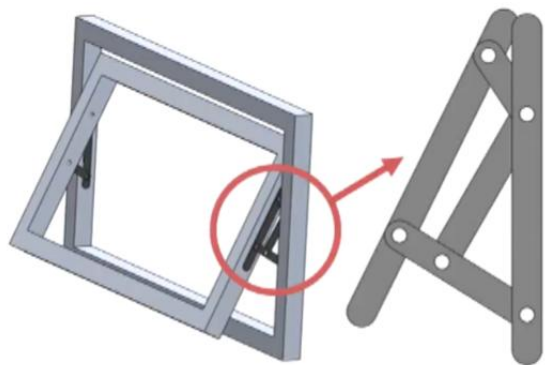
1. 机器人与机构学基础

1.3. 自由度与约束

自由度与约束的计算方法

- 注意：复合铰链带来的自由度计算问题

定义：两个以上的构件在同一处以转动副联接所构成的运动副
k个构件组成的复合铰链，有(k-1)个转动副



$$F = 3(l - 1) - 2P_1 - P_2$$
$$F = 3(6 - 1) - 2 \times 6 - 0 = 3$$
$$F = 3(6 - 1) - 2 \times 7 - 0 = 1$$

1. 机器人与机构学基础

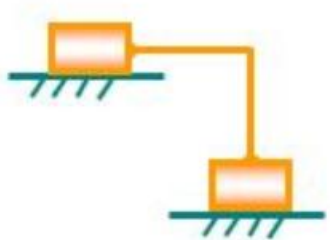
1.3. 自由度与约束

自由度与约束的计算方法

- 注意：虚约束带来的自由度计算问题：

定义： 机构中不起独立限制作用的重复约束，计算具有虚约束的机构的自由度时，应先将机构中引入虚约束的运动副或运动链部分除去

e.g. 两构件构成多个运动副



两构件构成多个
导路平行的移动副



两构件构成多个
轴线重合的转动副

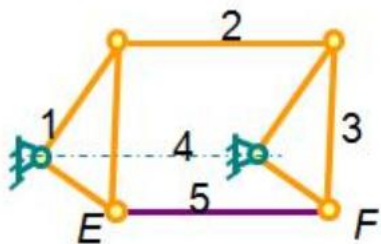
1. 机器人与机构学基础

1.3. 自由度与约束

自由度与约束的计算方法

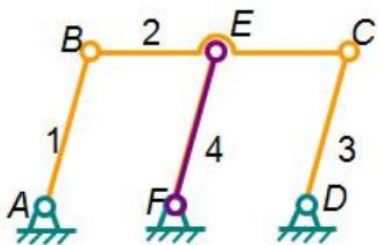
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未去掉虚约束时： $F=3(l-1)-2PL-PH=3*(5-1)-2*6=0$

去掉虚约束后 $F=3n-2PL-PH=3*(4-1)-2*4=1$



未去掉虚约束时： $F=3(l-1)-2PL-PH=3*(5-1)-2*6=0$

去掉虚约束后： $F=3(l-1)-2PL-PH=3*(4-1)-2*4=1$

平行四边形机构
构件2和4在E点轨迹重合

1.4. 常用传动机构

1. 机器人与机构学基础

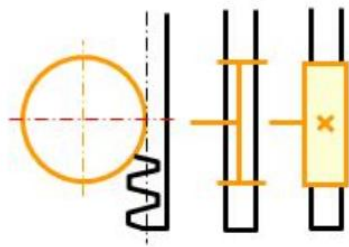
1.4. 常用传动机构

常用传动机构示例 (Transmission Mechanisms)

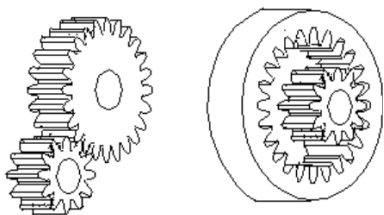
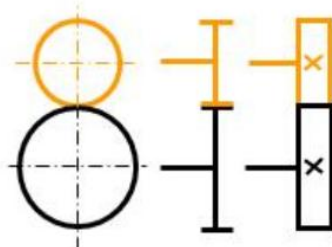
由构件的工作原理决定:

- 齿轮/齿系机构(Gear/Gear train)
- 凸轮和从动机构(Cam and follower)
- 带传动/链传动(Belt/Chain)

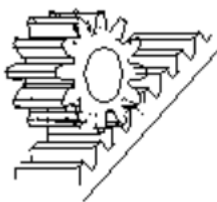
齿轮
齿条
传动



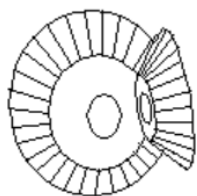
外啮合
圆柱
齿轮
传动



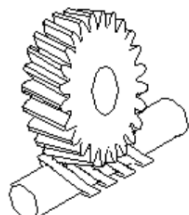
正齿轮、内齿轮



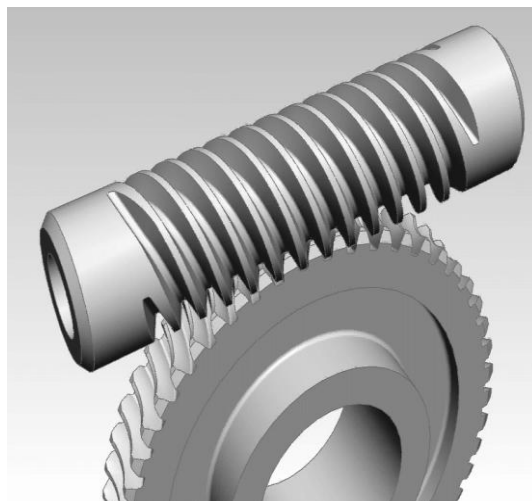
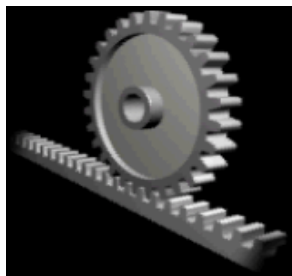
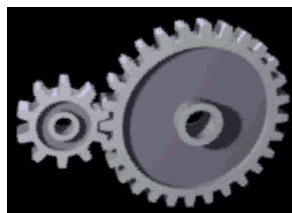
齿轮、齿条



直齿锥齿轮



涡轮、蜗杆



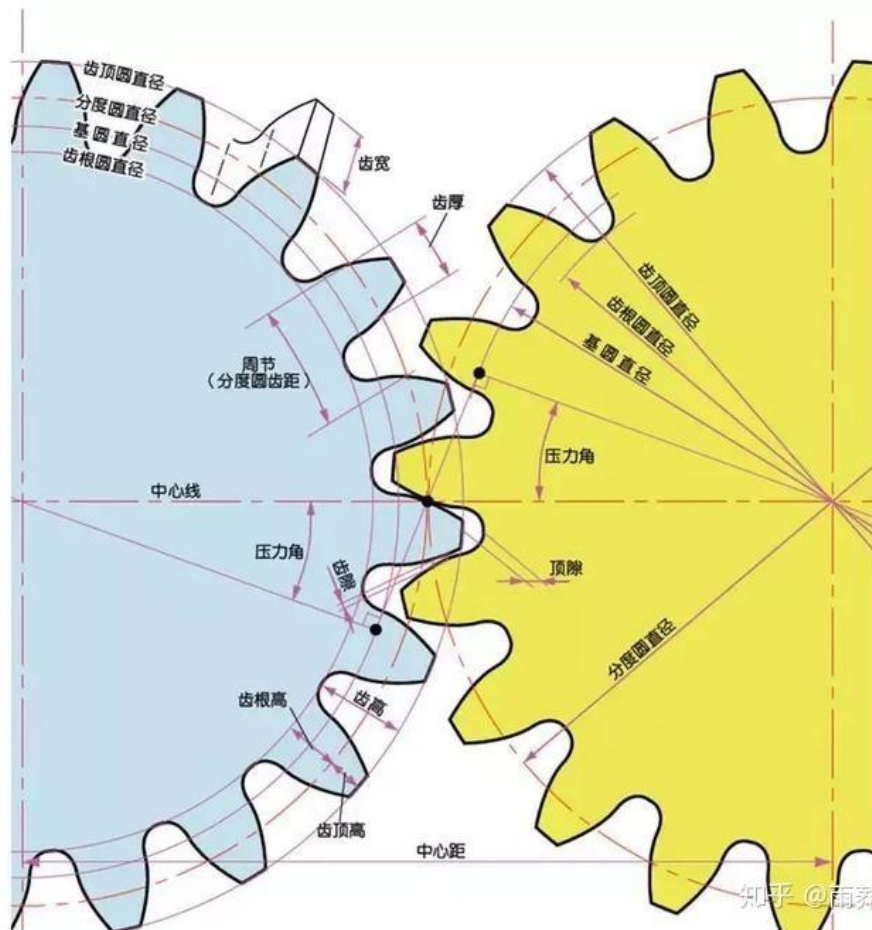
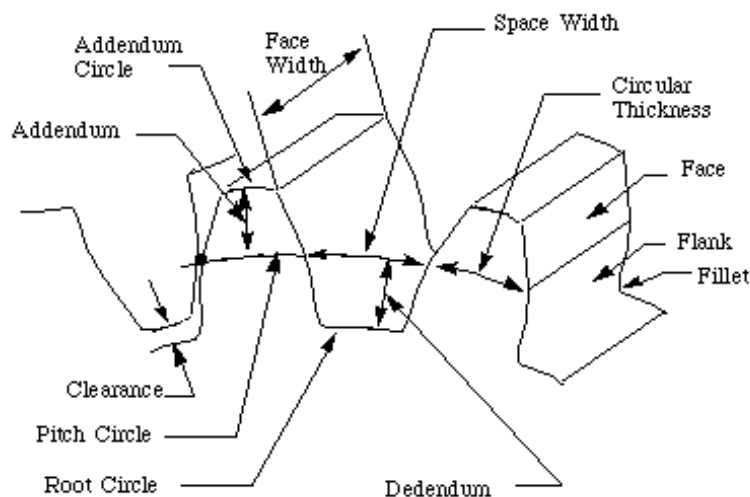
1. 机器人与机构学基础

1.4. 常用传动机构

常用传动机构示例 (Transmission Mechanisms)

由构件的工作原理决定:

- 齿轮/齿系机构(Gear/Gear train)
- 凸轮和从动机构(Cam and follower)
- 带传动/链传动(Belt/Chain)



1. 机器人与机构学基础

1.4. 常用传动机构

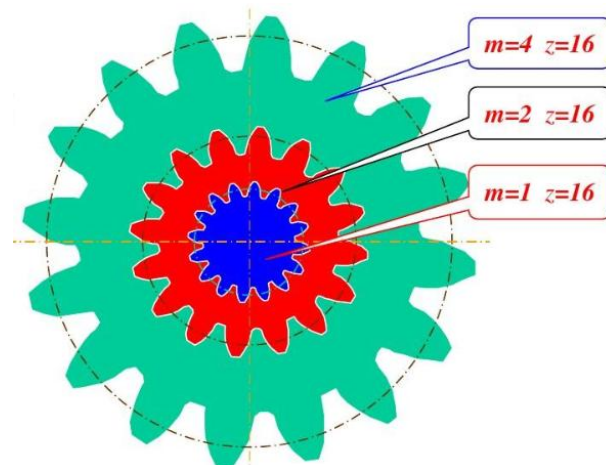
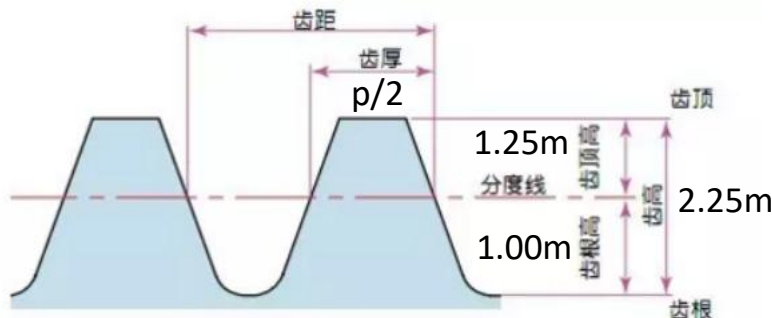
常用传动机构示例 (Transmission Mechanisms)

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① 模数(m)：表示齿轮大小的单位，模数乘以圆周率即可得到齿距(p)。

② 齿数(z)、分度圆周长： $\pi d = zp$ $m = p/\pi$
 $d = mz$, $r = mz/2$



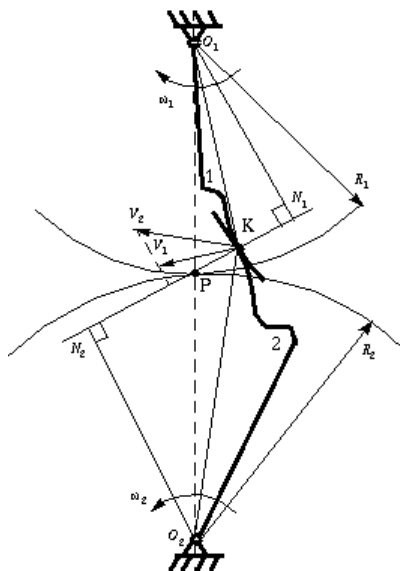
1. 机器人与机构学基础

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- 带传动/链传动(Belt/Chain)



$$O_1N_1 \cdot \omega_1 = O_2N_2 \cdot \omega_2$$

$$\frac{\omega_1}{\omega_2} = \frac{O_2N_2}{O_1N_1}$$

$$\Delta O_1N_1P \sim \Delta O_2N_2P$$

$$\frac{\omega_1}{\omega_2} = \frac{O_2P}{O_1P}$$

$$F_1 = \frac{T_1}{O_1P}$$

$$F_2 = \frac{T_2}{O_2P}$$

$$\frac{T_2}{T_1} = \frac{O_2P}{O_1P}$$

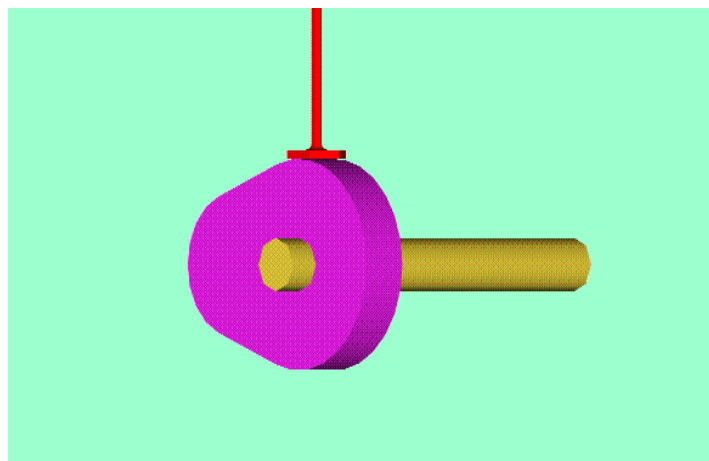
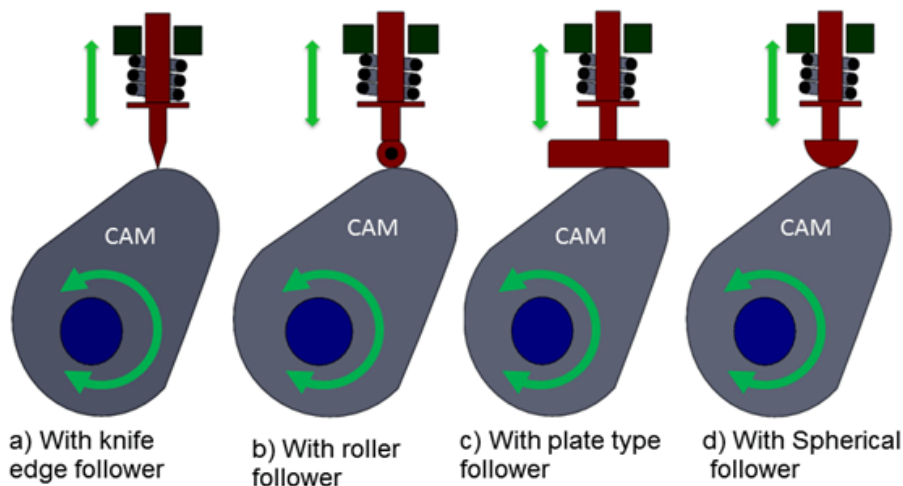
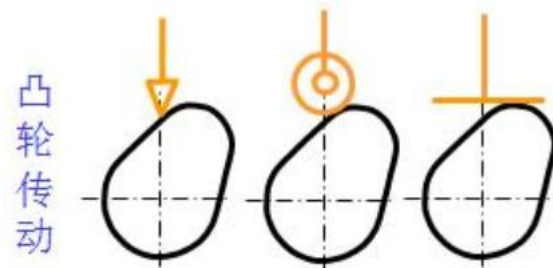
1. 机器人与机构学基础

1.4. 常用传动机构

常用传动机构示例 (Transmission Mechanisms)

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- 齿轮/齿系机构(Gear/Gear train)
- **凸轮和从动机构(Cam and follower)**
- 带传动/链传动(Belt/Chain)



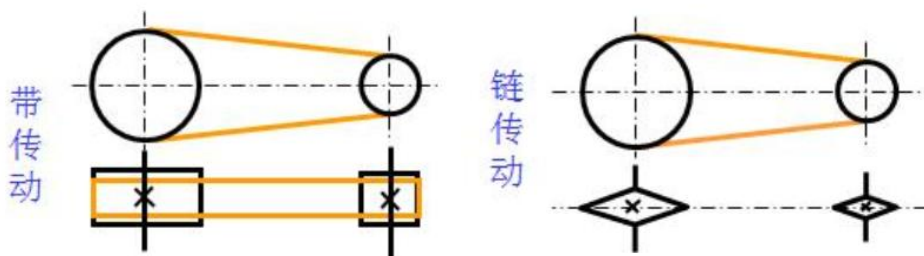
1. 机器人与机构学基础

1.4. 常用传动机构

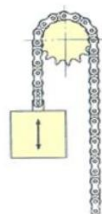
常用传动机构示例 (Transmission Mechanisms)

由构件的工作原理决定:

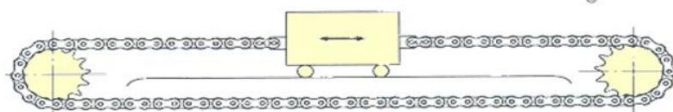
- 齿轮/齿系机构(Gear/Gear train)
- 凸轮和从动机构(Cam and follower)
- **带传动/链传动(Belt/Chain)**



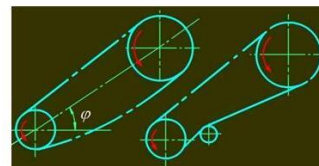
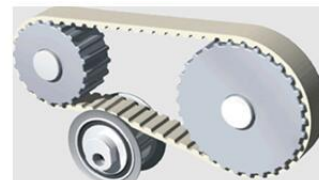
传动链



起重链



输送链



带传动: 主动带轮、从动带轮、带
同步带: 带齿与轮齿的相互啮合传动

链传动: 主动链轮、从动链轮、链条
链轮轮齿与链节的啮合传递运动和动力

张紧轮的使用

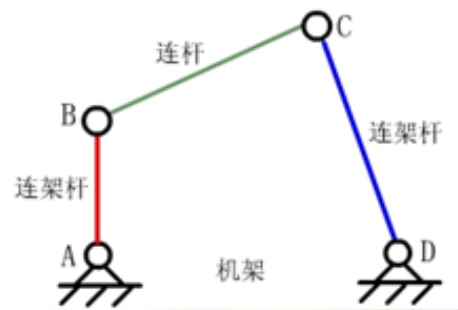
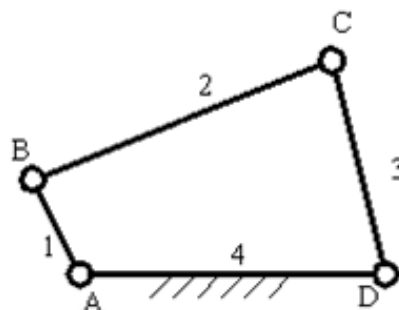
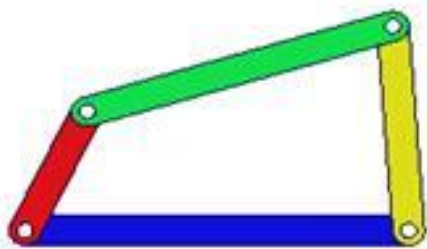
1.5. 平面连杆机构

1. 机器人与机构学基础

1.5. 平面连杆机构

(1) 平面四杆机构 (4-bar Linkage)

- 铰链四连杆机构是有四个杆件通过铰链连接而成的传动机构，简称四杆机构



- 机架 (Frame)：机构的固定构件，如杆4
- 连杆 (Coupler link)：不直接与机架连接的构件，如杆2
- 连架杆 (Side Links)：与机架用转动副相连接的构件

连架杆可分为：

- 曲柄 (Crank)：能绕机架作整周转动的连架杆
- 摇杆 (Rocker)：只能绕机架作小于 360° 的某一角度摆动的连架杆

1. 机器人与机构学基础

1.5. 平面连杆机构

(1) 平面四杆机构 (4-bar Linkage)

Grashof's theorem (葛氏定理)

s = length of shortest bar

l = length of longest bar

p, q = lengths of intermediate bar

$l + s$ vers. $p + q$ Shortest Bar

$<$

Frame

Double-crank

$<$

Side

Rocker-crank

$=$

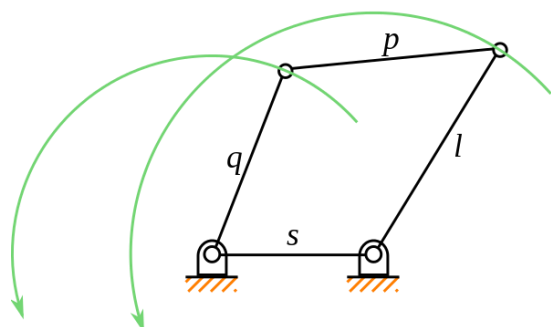
Any

Change point

$>$

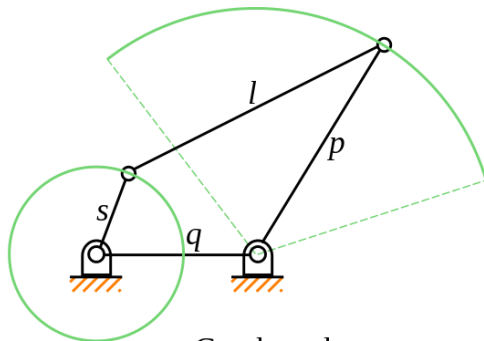
Any

Double-rocker

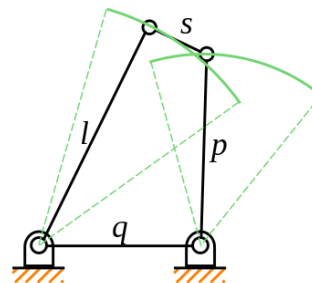


full revolution
both links

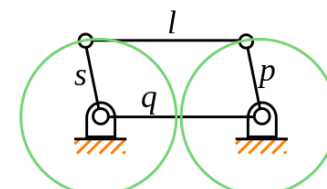
Drag-link
 $s + l < p + q$
(continuous motion)



Crank-rocker
 $s + l < p + q$
(continuous motion)



Double-rocker
 $s + l > p + q$
(no continuous motion)



Parallelogram linkage
 $s + l = p + q$
(continuous motion)

1. 机器人与机构学基础

1.5. 平面连杆机构

(1) 平面四杆机构 (4-bar Linkage)

Checking for Grashof Condition

$$\text{if : } S + L \leq P + Q$$

- This is called a **Grashof Linkage** and
- ***at least one link will be capable of making a full revolution*** with respect to the ground plane.
- This is also called a **Class I kinematic chain.**

$$\text{if : } S + L > P + Q$$

- The **linkage is non-Grashof** and
- ***no link will be capable of a complete revolution*** relative to any other link.*
- This is a **Class II kinematic chain.**

1. 机器人与机构学基础

1.5. 平面连杆机构

(1) 平面四杆机构 (4-bar Linkage)

TABLE 2-4 Barker's Complete Classification of Planar Fourbar Mechanisms

Adapted from ref. (10). s = shortest link, l = longest link, Gxxx = Grashof, RRRx = non-Grashof, Sxx = Special case

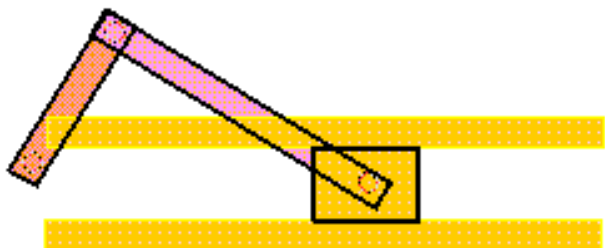
Type	$s + l$ vs. $p + q$	Inversion	Class	Barker's Designation	Code	Also Known As
1	<	$L_1 = s = \text{ground}$	I-1	Grashof crank-crank-crank	GCCC	double-crank
2	<	$L_2 = s = \text{input}$	I-2	Grashof crank-rocker-rocker	GCRR	crank-rocker
3	<	$L_3 = s = \text{coupler}$	I-3	Grashof rocker-crank-rocker	GRCR	double-rocker
4	<	$L_4 = s = \text{output}$	I-4	Grashof rocker-rocker-crank	GRRC	rocker-crank
5	>	$L_1 = l = \text{ground}$	II-1	Class 1 rocker-rocker-rocker	RRR1	triple-rocker
6	>	$L_2 = l = \text{input}$	II-2	Class 2 rocker-rocker-rocker	RRR2	triple-rocker
7	>	$L_3 = l = \text{coupler}$	II-3	Class 3 rocker-rocker-rocker	RRR3	triple-rocker
8	>	$L_4 = l = \text{output}$	II-4	Class 4 rocker-rocker-rocker	RRR4	triple-rocker
9	=	$L_1 = s = \text{ground}$	III-1	change point crank-crank-crank	SCCC	SC [*] double-crank
10	=	$L_2 = s = \text{input}$	III-2	change point crank-rocker-rocker	SCRR	SC crank-rocker
11	=	$L_3 = s = \text{coupler}$	III-3	change point rocker-crank-rocker	SRCR	SC double-rocker
12	=	$L_4 = s = \text{output}$	III-4	change point rocker-rocker-crank	SRRC	SC rocker-crank
13	=	two equal pairs	III-5	double change point	S2X	parallelogram or deltoid
14	=	$L_1 = L_2 = L_3 = L_4$	III-6	triple change point	S3X	square

* SC = special case.

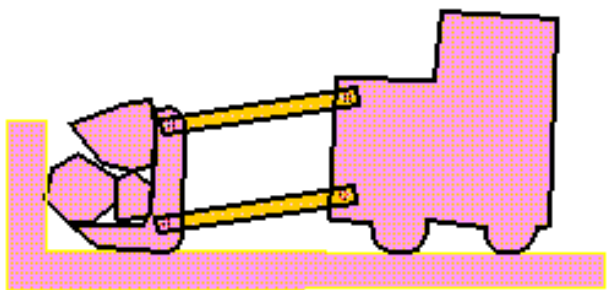
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1.5. 平面连杆机构

(1) 平面四杆机构 (4-bar Linkage)



通过使一个或多个链接的长度变成无限长，四杆机构具有一些特殊的配置。**滑块-曲柄（或曲柄-滑块）机构**是一种特殊的四杆机构，滑块取代了无限长的输出连杆。

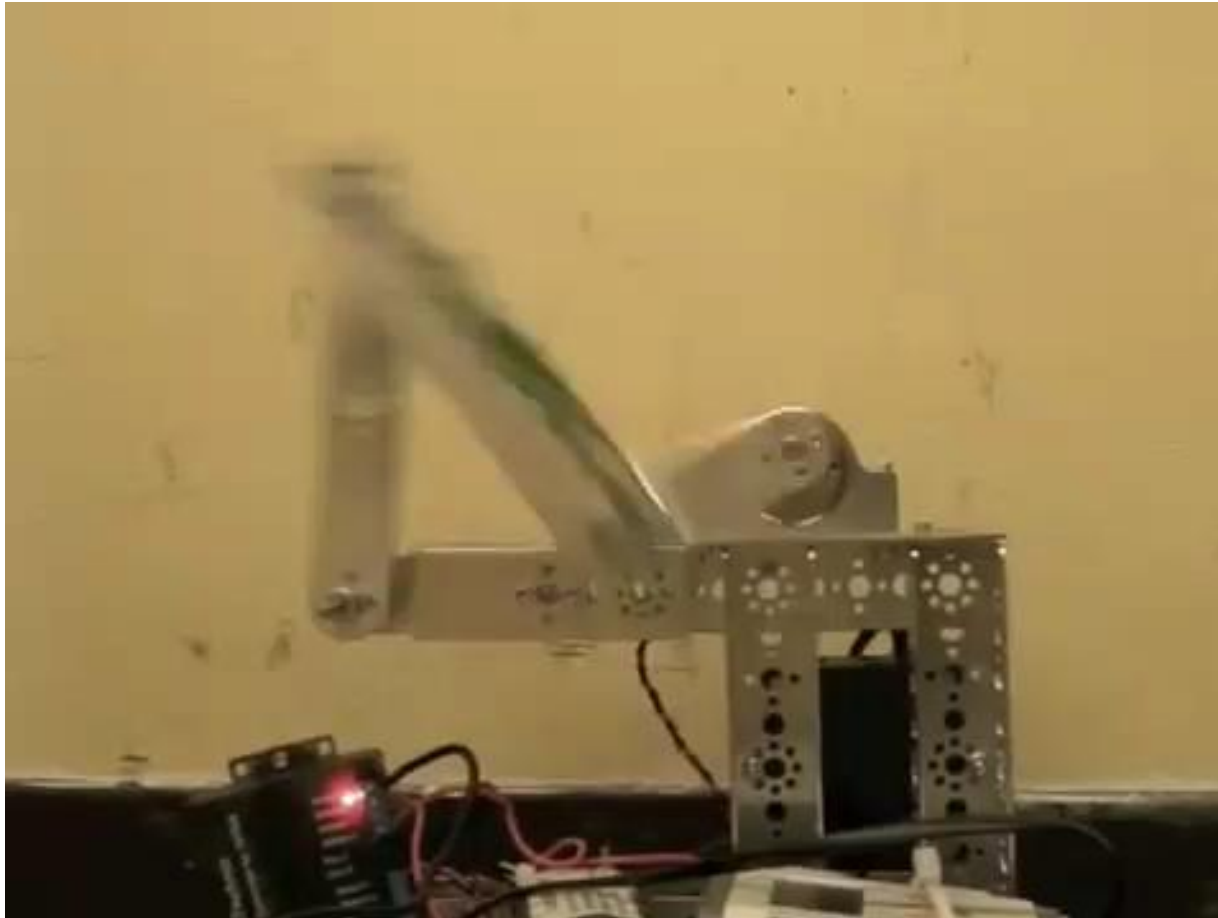


在**平行四边形四杆机构**中，铲斗的方向在运动过程中不会改变，铲斗在升高和降低时不会旋转

1. 机器人与机构学基础

1.5. 平面连杆机构

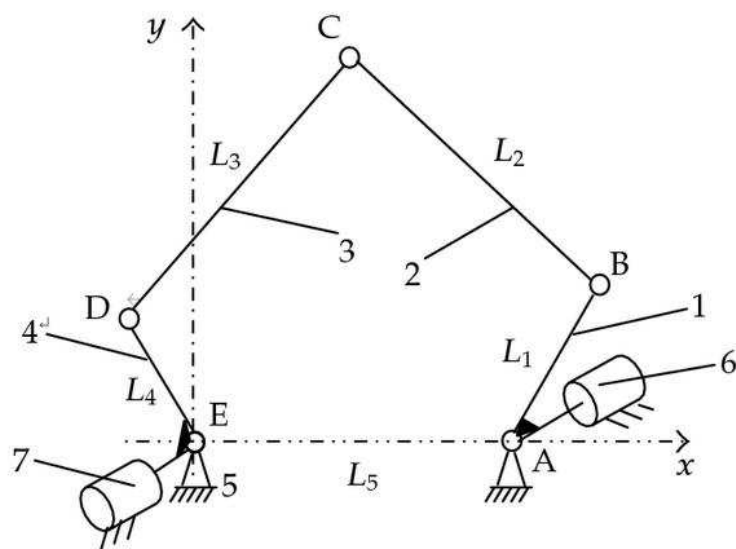
(1) 平面四杆机构 (4-bar Linkage)



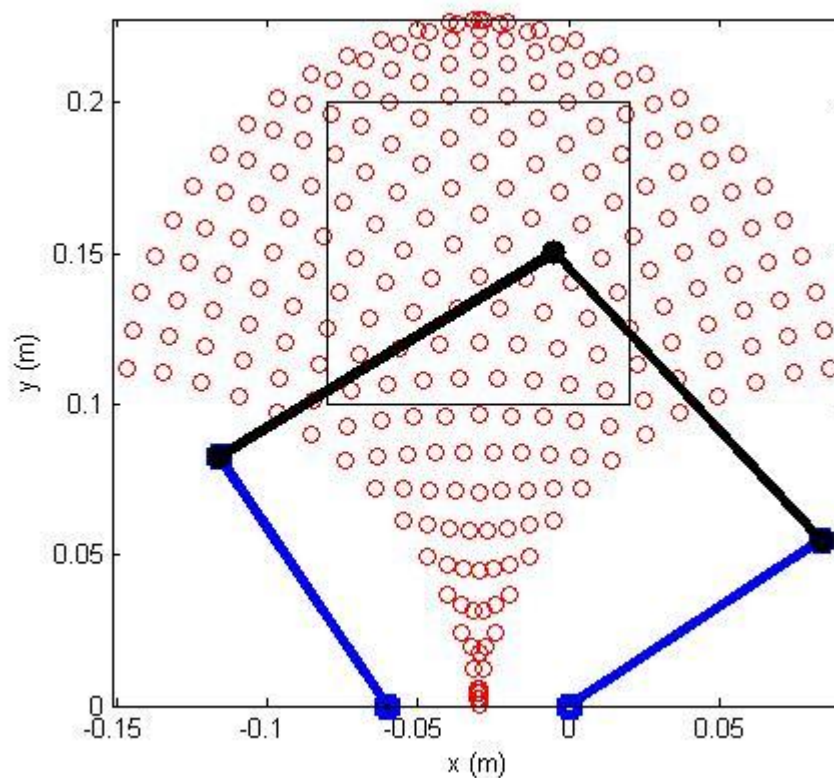
1. 机器人与机构学基础

1.5. 平面连杆机构

(2) 平面五杆机构 (5-bar Linkage)



- | | |
|------------|------------|
| 1. Crank | 5. Frame |
| 2. Coupler | 6. Motor 1 |
| 3. Coupler | 7. Motor 2 |
| 4. Crank | |

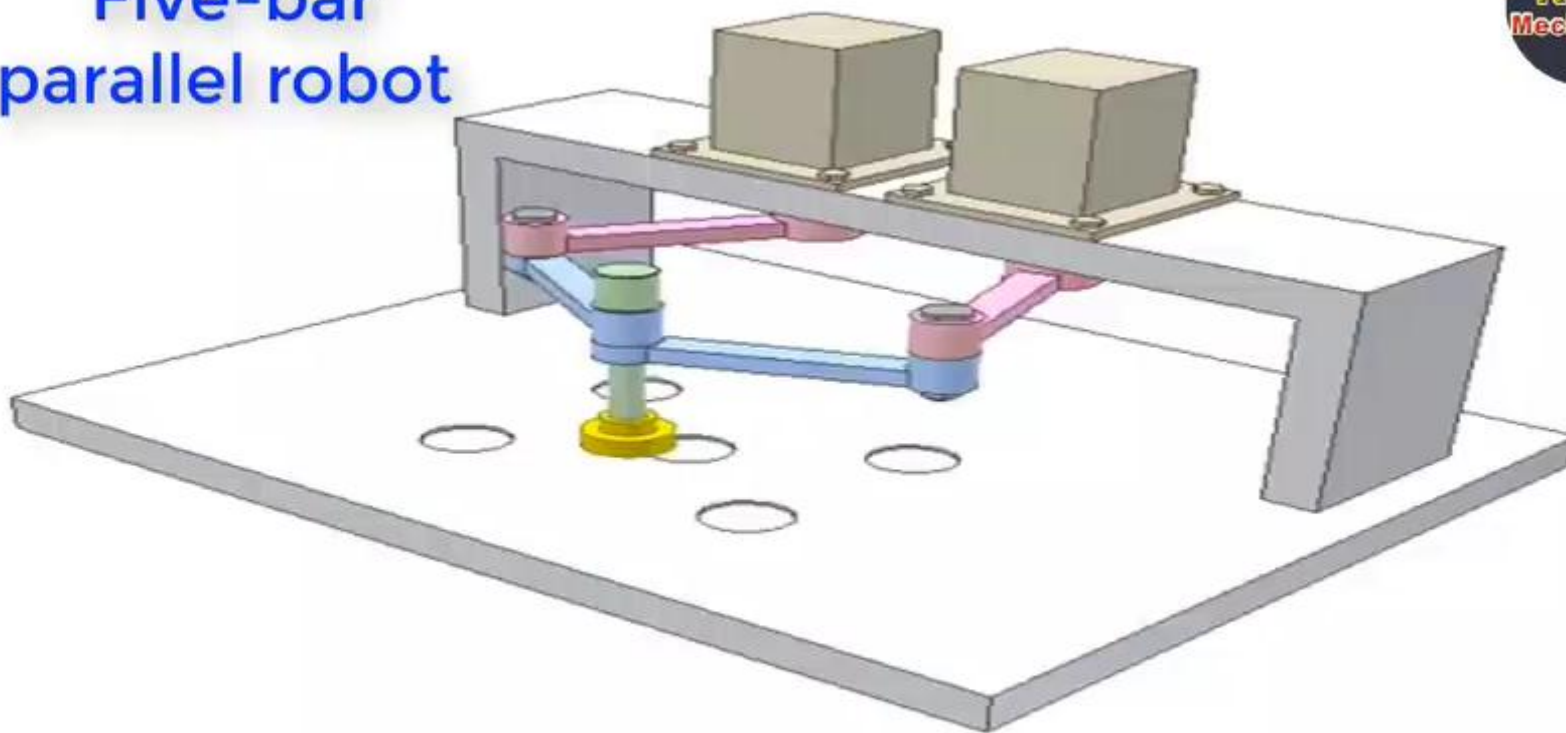


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1.5. 平面连杆机构

(2) 平面五杆机构 (5-bar Linkage)

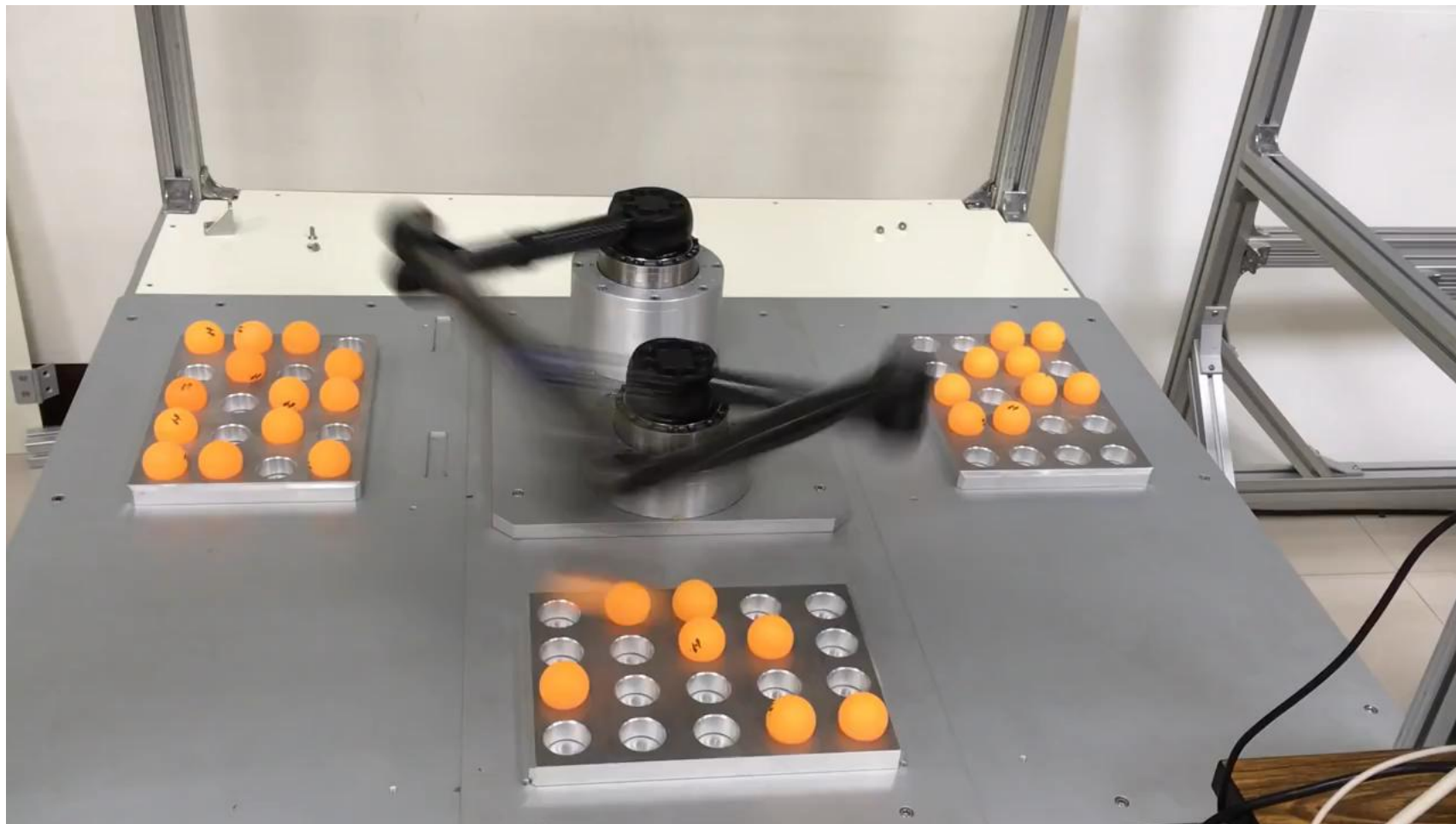
Five-bar
parallel robot



1. 机器人与机构学基础

1.5. 平面连杆机构

(2) 平面五杆机构 (5-bar Linkage)



1. 机器人与机构学基础

1.6. 机构分析工具

(1) Linkage

<https://blog.rectorsquid.com/linkage-mechanism-designer-and-simulator/>

Linkage Mechanism Designer and Simulator

Mac Version Coming Soon! (in a year or two;)

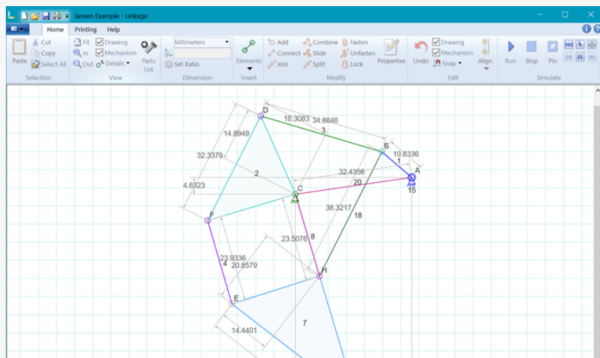
Linkage is a free computer-aided design program for Microsoft Windows used for prototyping of mechanical linkages. The mechanism is edited and animated in the same window allowing for quick analysis and modification while working on a design. It is simplistic for a CAD program but that is the intent.

The current version is 3.16.14

➤ Download Linkage files here including documentation and beta software.

➤ Click here for Frequently Asked Questions.

If you have winget installed, you can run "winget install linkage" at a command prompt to install Linkage.



Linkage 3.16

User's Guide

David Rector
Wednesday, August 03, 2022

Simple Cam Tutorial

Splines can be dropped into the mechanism from the right-click element gallery. A closed spline can be fastened to a link and used as a cam with a sliding connector acting as the cam follower.

This tutorial is less detailed than the first mechanism tutorial. Refer to that tutorial if any of these steps seem confusing.

1. Add a link on a rotating input connector using the popup element gallery.



Figure 22. Cam Tutorial Simple Rotating Link

2. Add a spline.



Figure 23. Cam Tutorial Newly Inserted Spline

3. Close the spline, move it to be near or around the link.
4. Shape the spline to the desired cam shape.

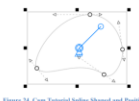


Figure 24. Cam Tutorial Spline Shaped and Positioned

1. 机器人与机构学基础

1.6. 机构分析工具

(2) DAS 2D/3D

DAS 2D/3D

Compliant mechanism analysis

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Matlab Files

Source files are available here. Choose this if you want to see or edit the source files. Some of the source files will only work with MATLAB 2016a or later.

Direct:[Download](#)

Online Installation

This file does not contain Matlab Run-Time files for installing DAS 2D/3D to your computer. Internet connection is necessary during installation. Choose this if you installed the software before.

Direct:[Download \(64 Bit\)](#)

<http://www.compliantanalysis.com/>

1. 机器人与机构学基础

1.6. 机构分析工具

(2) DAS 2D/3D



Design, Analysis and Synthesis of Compliant Mechanisms



DAS 2D/3D is a software tool capable of analysis and synthesis of compliant mechanisms.

- DAS 2D can perform
 - Kinematic Analysis and Synthesis
 - Static Analysis and Synthesisfor rigid and compliant planar mechanisms.



Kinematics



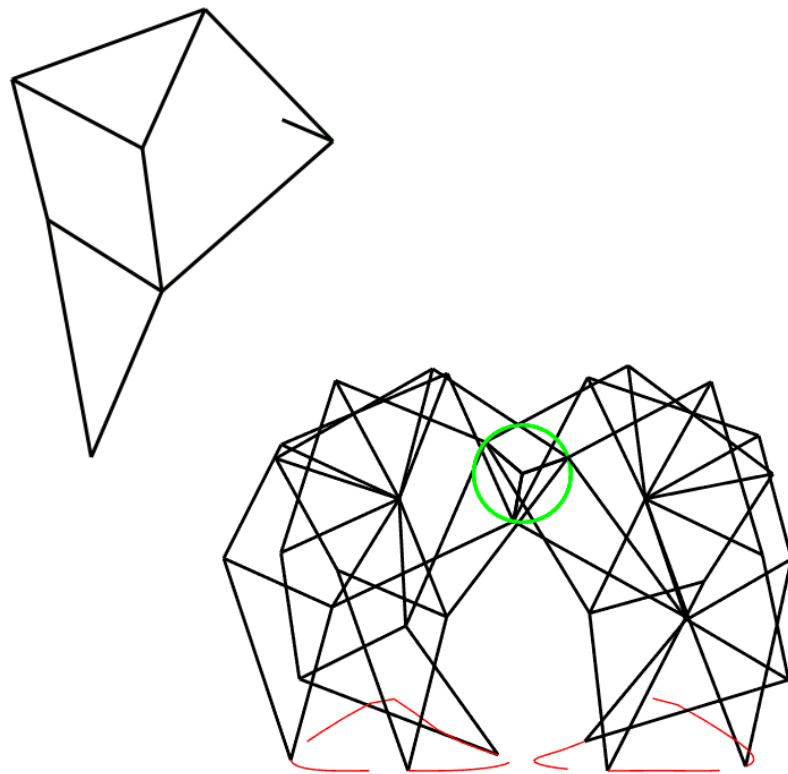
Statics

- DAS 3D will perform
 - Static Analysis and Synthesisfor flexure mechanisms.

1. 机器人与机构学基础

➤ 第一章作业

2. 查阅资料，使用DAS2D/3D绘制Jansen连杆机构的运动简图（请自行设定杆件尺寸），分析杆件及运动副数量、关键构件的运动轨迹及该机构自由度。



1. 机器人与机构学基础

➤ 第一章作业

3. 查阅资料，使用DAS2D/Linkage绘制下述挖掘机机构的运动简图（请自行设定杆件尺寸），分析杆件及运动副数量、关键构件的运动轨迹及该机构自由度。

